

Busca em profundidade

Inicialização

$t \leftarrow 0$ -- t é o relógio ou tempo global

para todo vértice v em $V(G)$ faça

$PE(v) \leftarrow 0$ -- $PE(v)$ é a profundidade de entrada de v

$PS(v) \leftarrow 0$ -- $PS(v)$ é a profundidade de saída de v

$pai(v) \leftarrow \text{null}$ -- ponteiros que definem a árvore de profundidade

Chamada Externa

escolher um vértice qualquer v em $V(G)$ -- este vértice é chamado **raiz da busca**
executar $P(v)$

Busca em profundidade

Procedimento recursivo de busca

procedimento $P(v)$

$t \leftarrow t + 1$

$PE(v) \leftarrow t$

para todo vértice w em $N(v)$ faça

se $PE(w) = 0$

então { $\begin{array}{l} \text{visitar aresta } vw \quad (\text{aresta “azul” da árvore de profundidade } T) \\ \text{pai}(w) \leftarrow v \quad (v \text{ é o pai de } w \text{ na árvore de profundidade } T) \\ \text{executar } P(w) \end{array}$

senão { $\begin{array}{l} \text{se } PS(w) = 0 \text{ e } w \neq \text{pai}(v) \quad (\text{se } w \text{ não saiu da busca e não é pai de } v) \\ \quad \text{então visitar aresta } vw \quad (\text{aresta “vermelha” de retorno}) \end{array}$

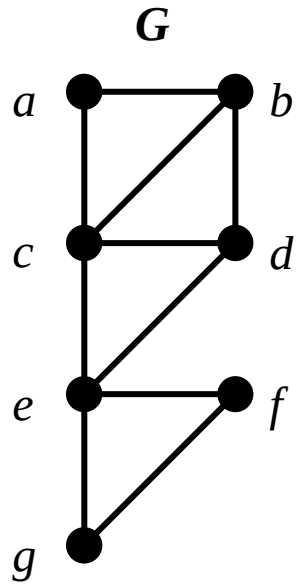
fim-para

$t \leftarrow t + 1$

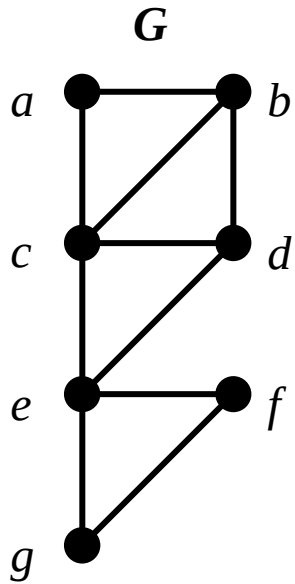
$PS(v) \leftarrow t$

fim-do-procedimento

Busca em profundidade

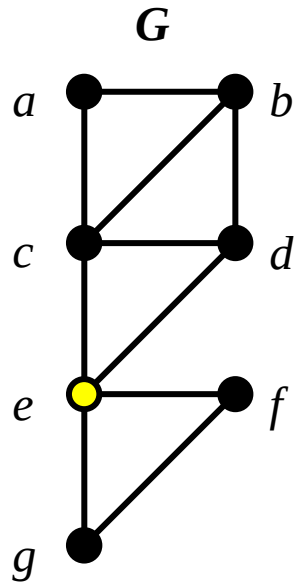


Busca em profundidade



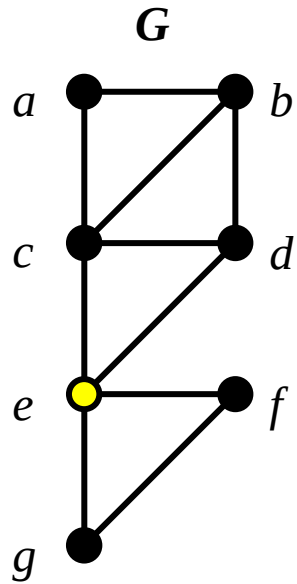
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	0	0	0	0	0	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	0	0	0	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

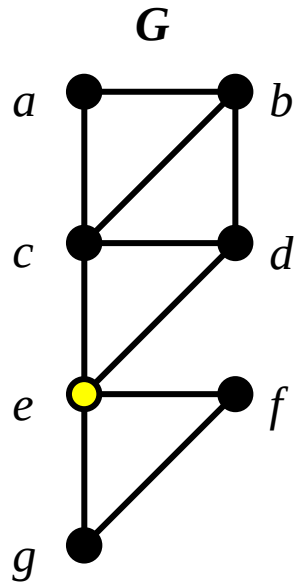
Busca em profundidade



$P(e)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	0	0	0	0	0	0	0
$PS(v)$	0	0	0	0	0	0	0

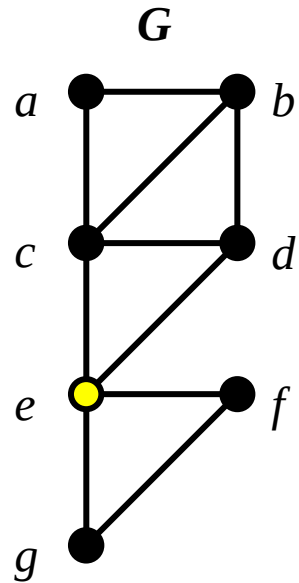
Busca em profundidade



$P(e)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	0	0	0	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



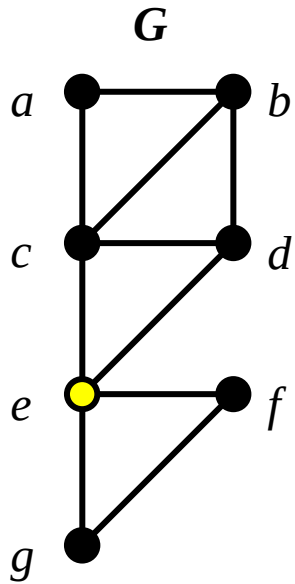
T

e ●

$P(e)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	0	0	0	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

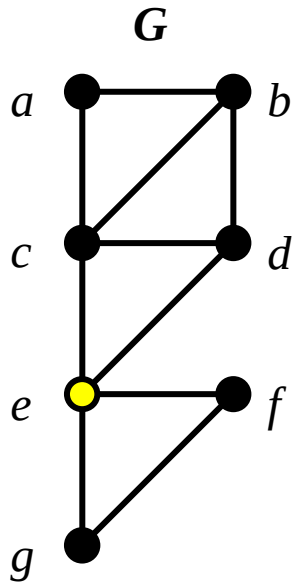
Busca em profundidade



$$P(e) \rightarrow N(e) = \{c, d, f, g\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	0	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

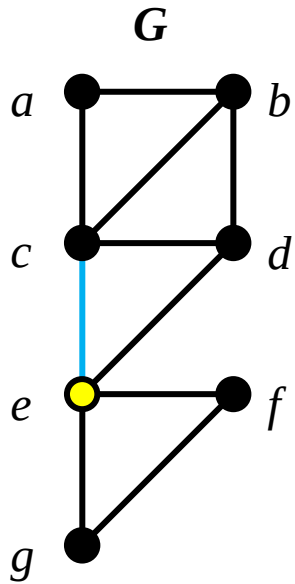
Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	0	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade

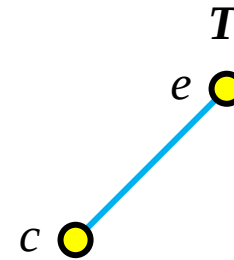
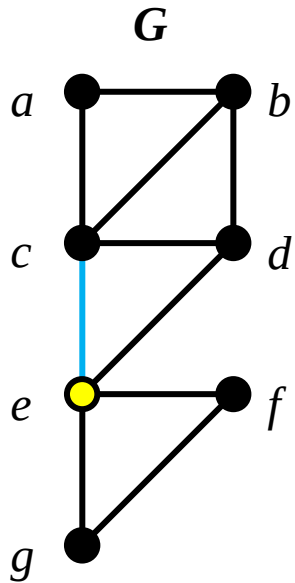


T
e ●

$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	0	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

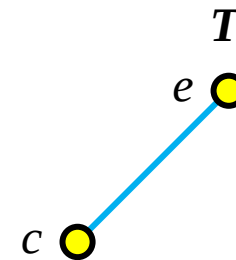
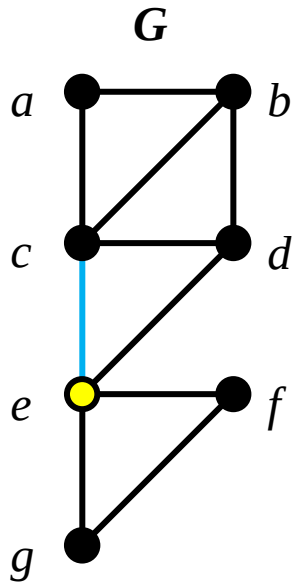
Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	0	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade

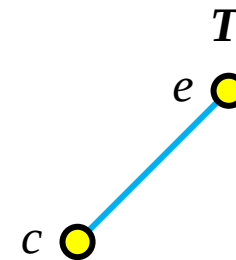
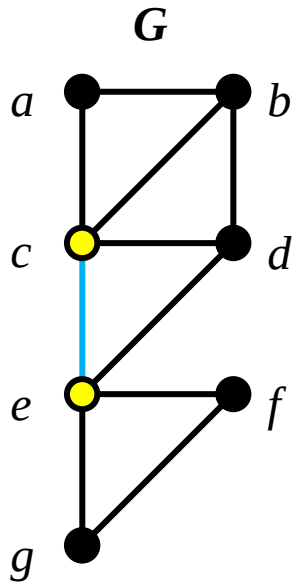


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c)$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	0	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade

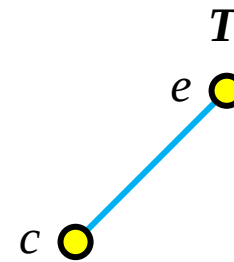
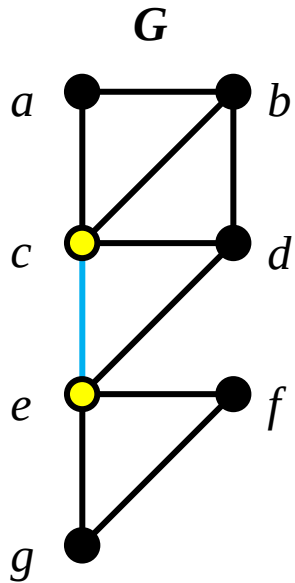


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c)$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	0	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade

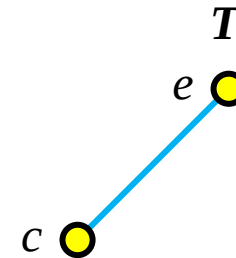
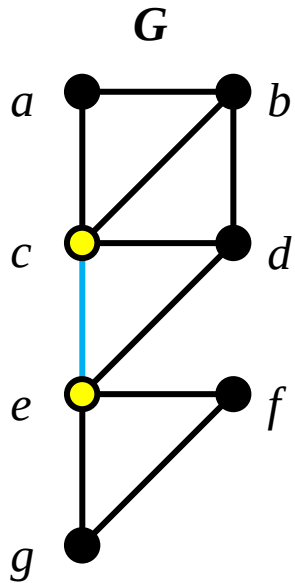


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c)$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade

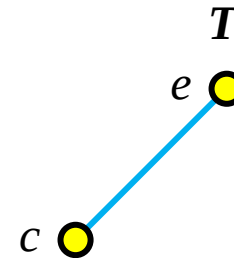
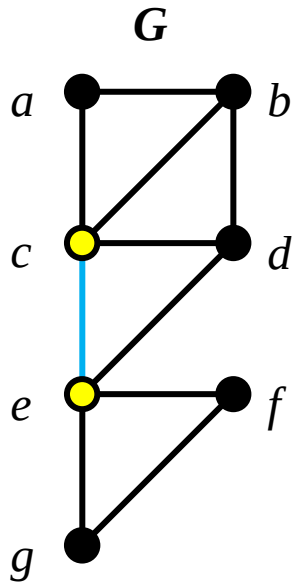


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c) \rightarrow N(c) = \{a, b, d, e\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade

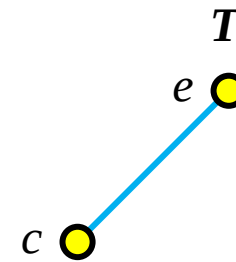
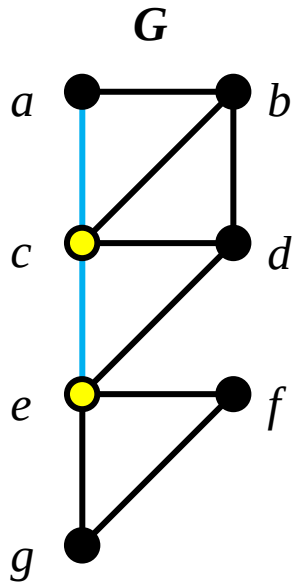


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade

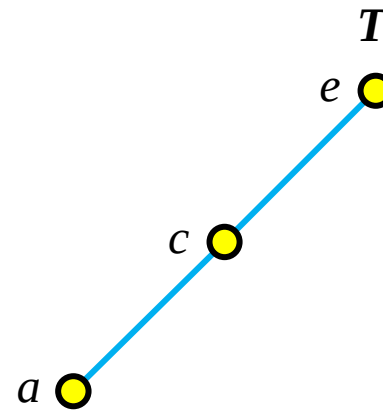
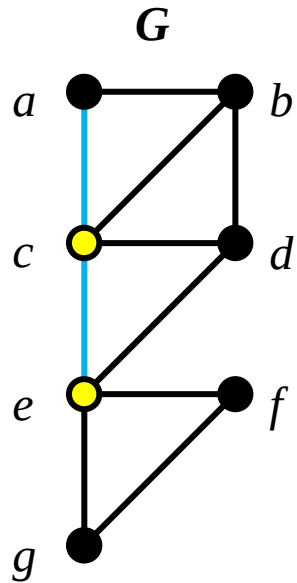


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade

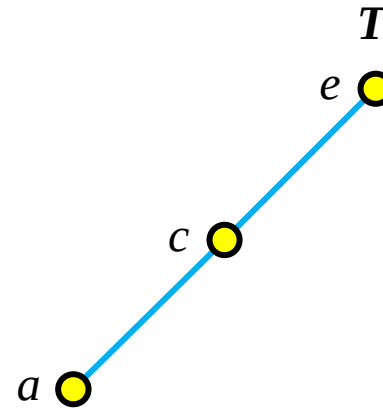
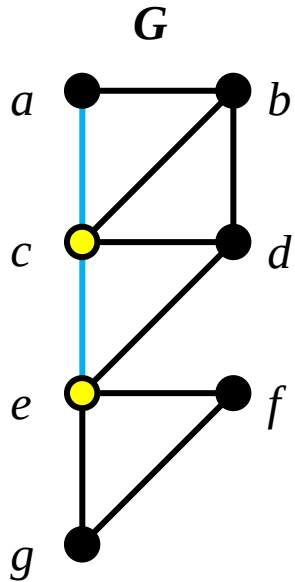


$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



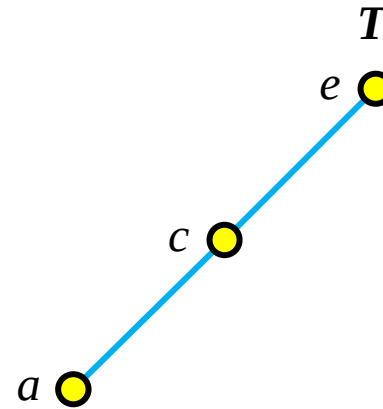
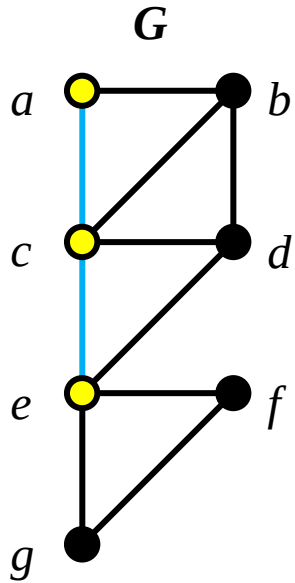
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	0	0	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



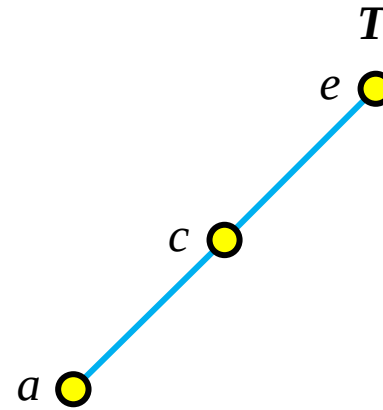
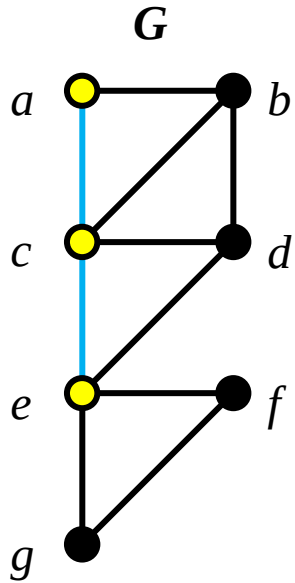
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	0	0	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade



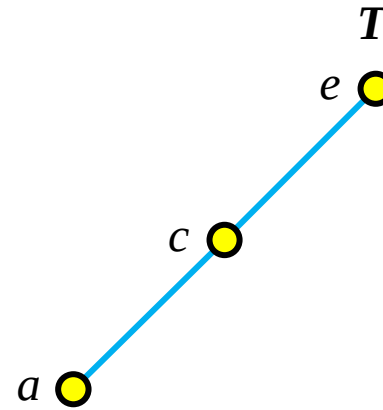
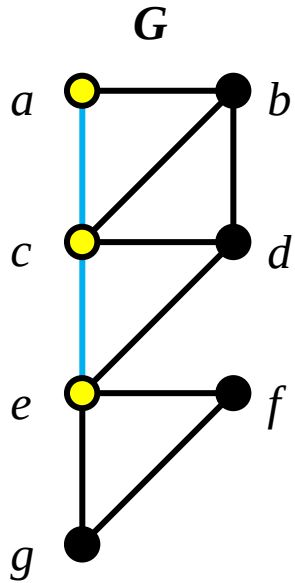
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	0	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



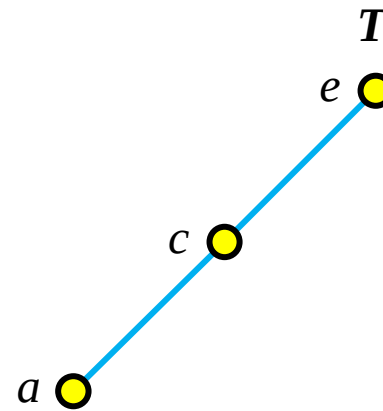
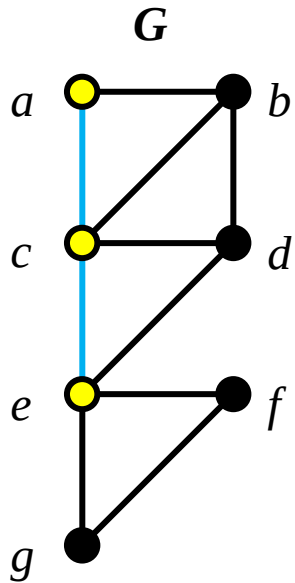
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{b, c\}$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	0	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



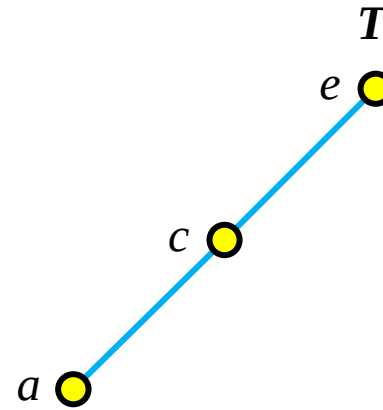
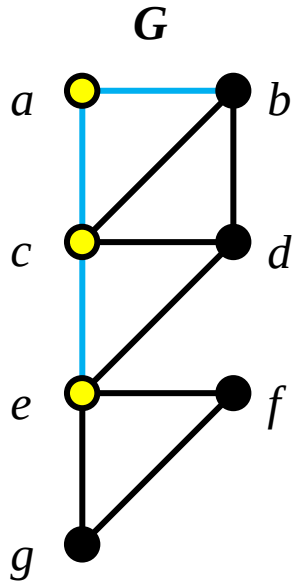
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	0	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade



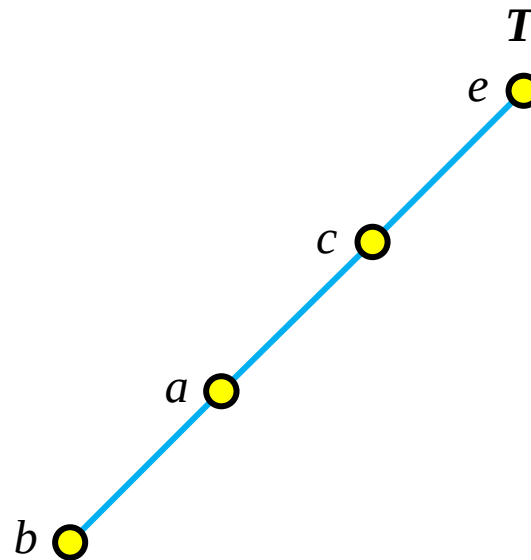
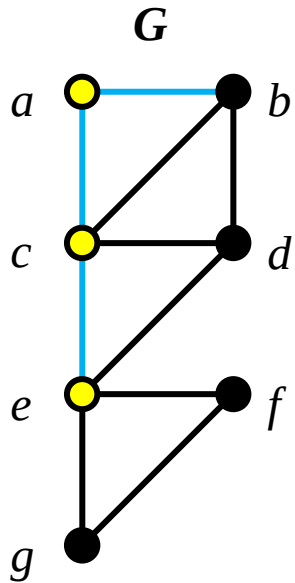
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	0	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



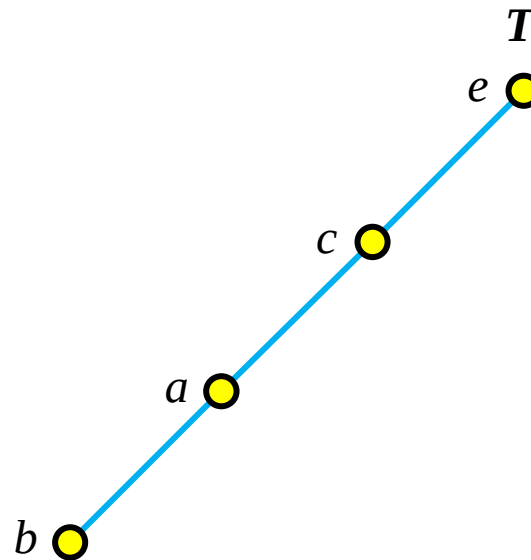
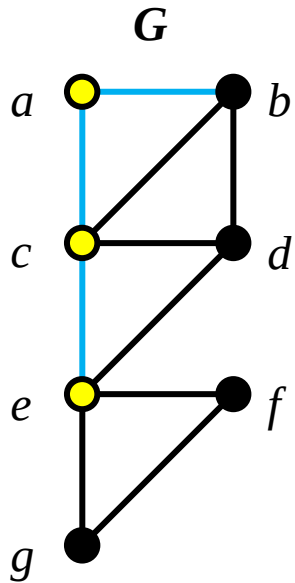
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	0	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

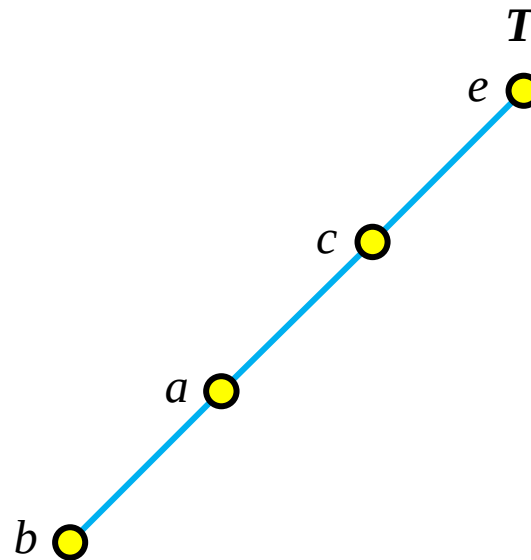
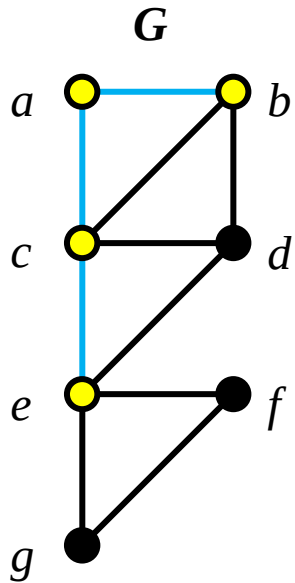
$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

$P(b)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	0	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

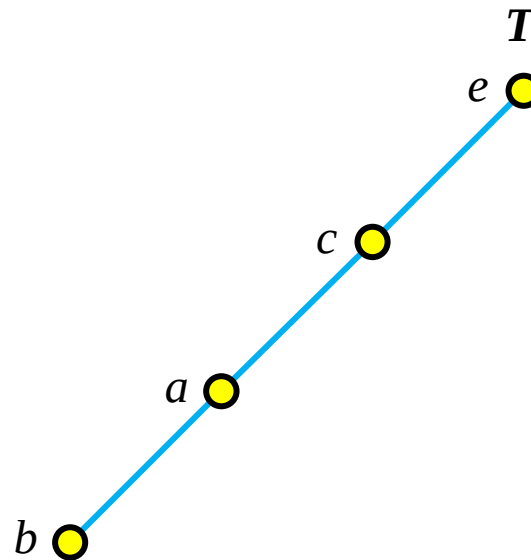
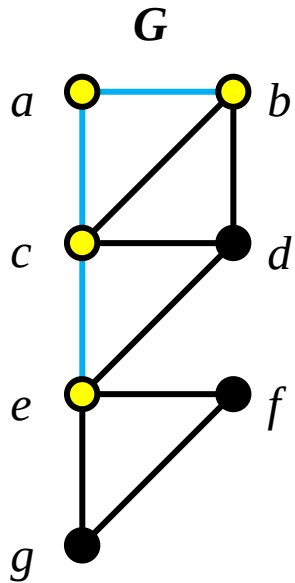
$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

$P(b)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	0	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

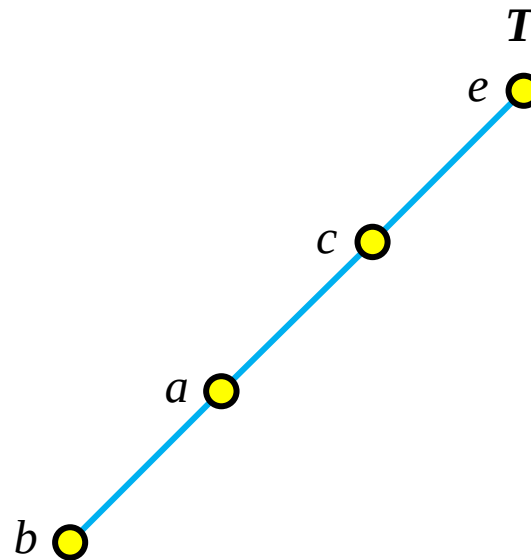
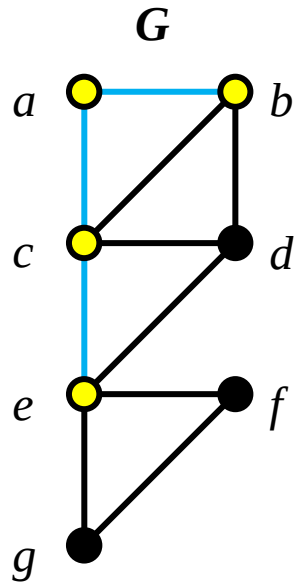
$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

$P(b)$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

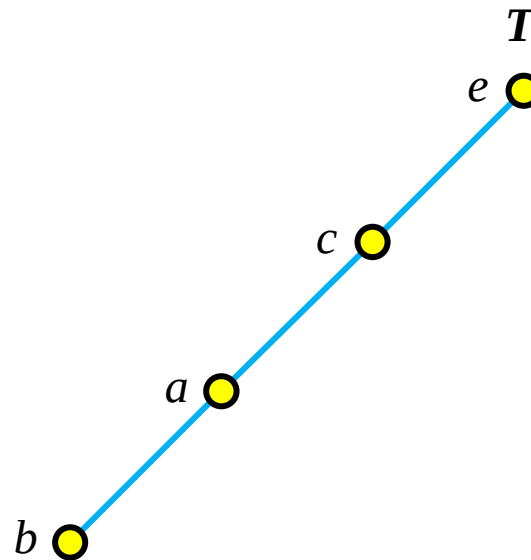
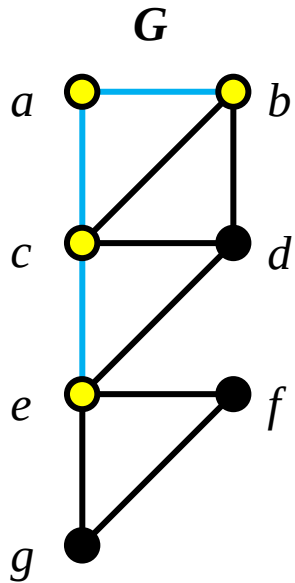
$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$$

$$P(b) \rightarrow N(b) = \{a, c, d\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

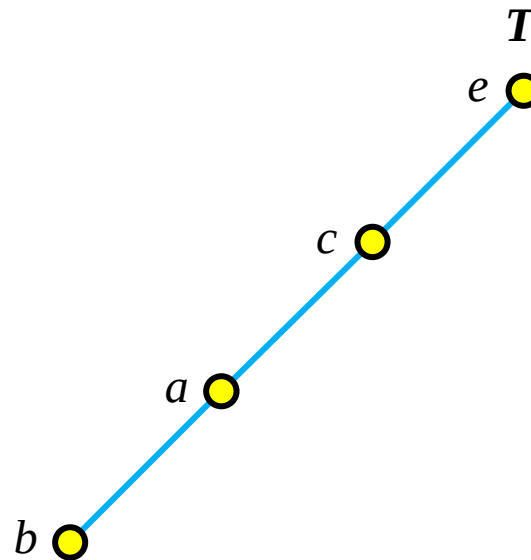
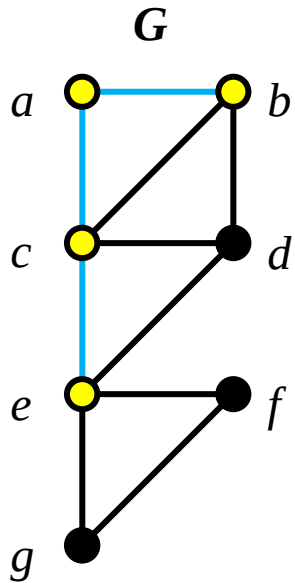
$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$$

$$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, c, d\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	0	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

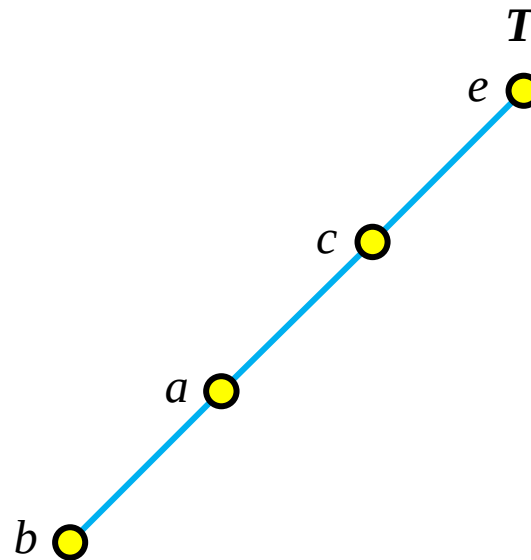
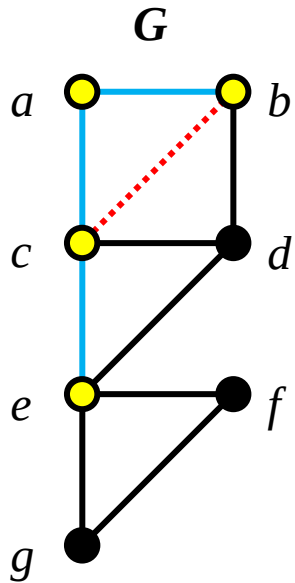
$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$$

$$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, d\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

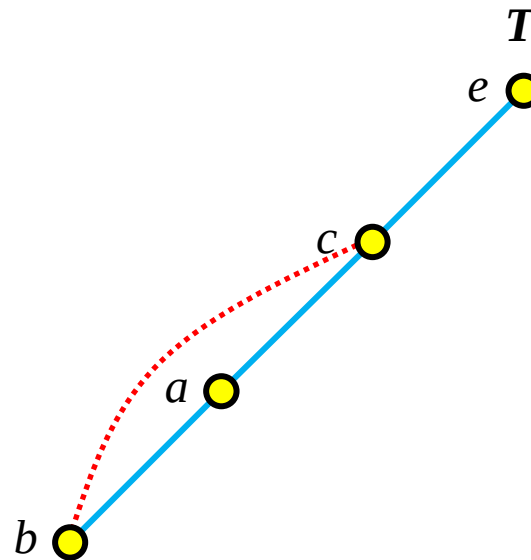
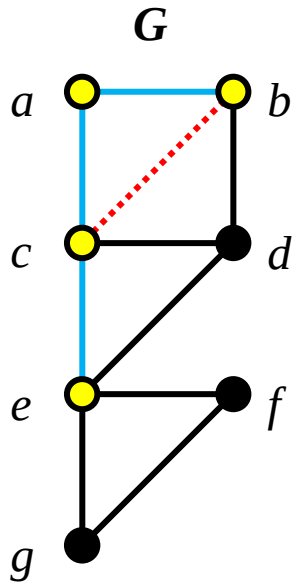
$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$$

$$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, d\}$$

<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$$

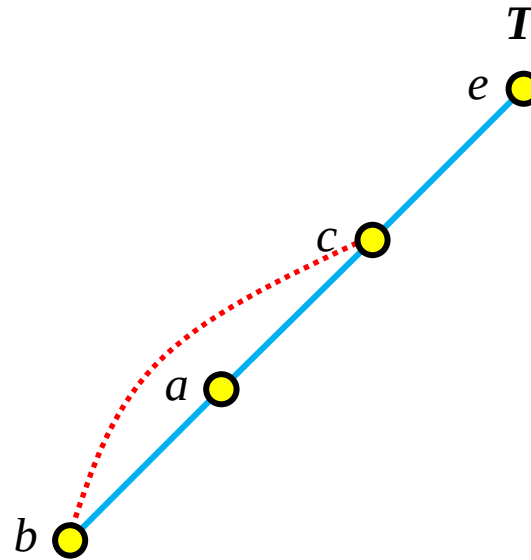
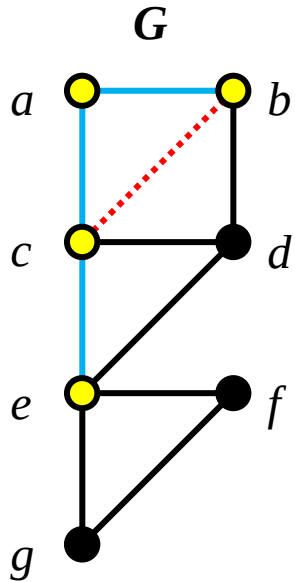
$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$$

$$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, d\}$$

v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

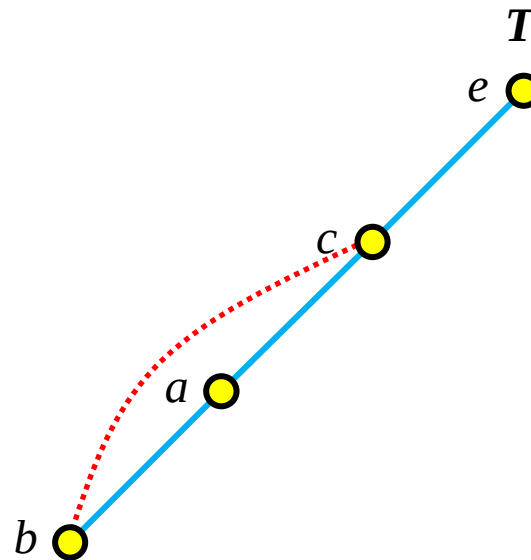
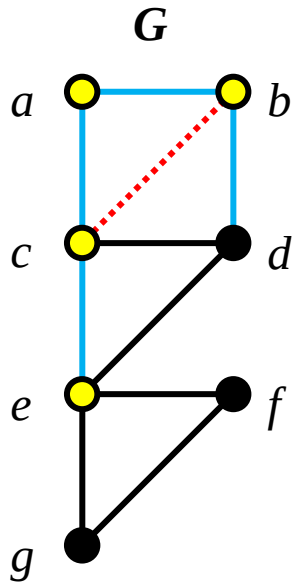
$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{c, d, f, g\}$$

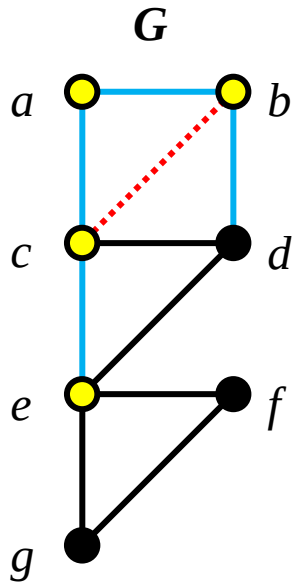
$$P(c) \rightarrow N(c) = \{a, b, d, e\}$$

$$P(a) \rightarrow N(a) = \{b, c\}$$

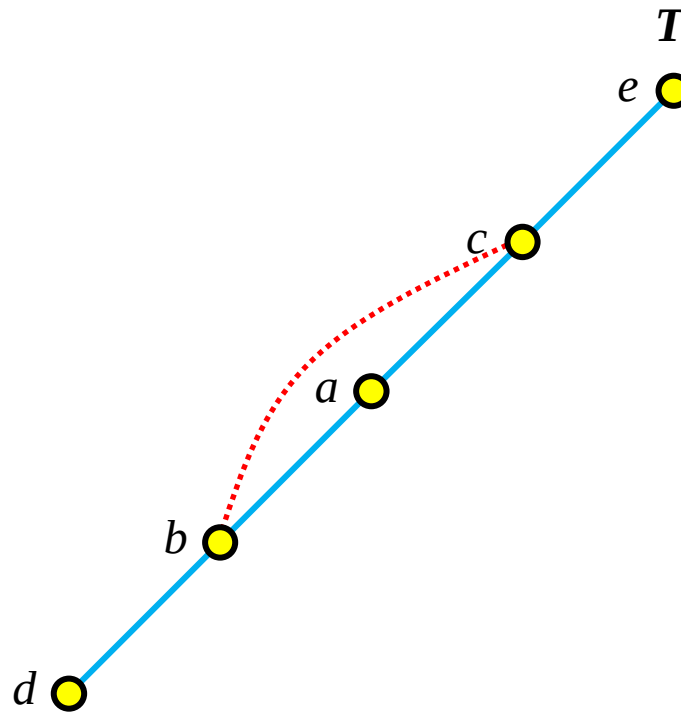
$$P(b) \rightarrow N(b) = \{a, c, d\}$$

v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade

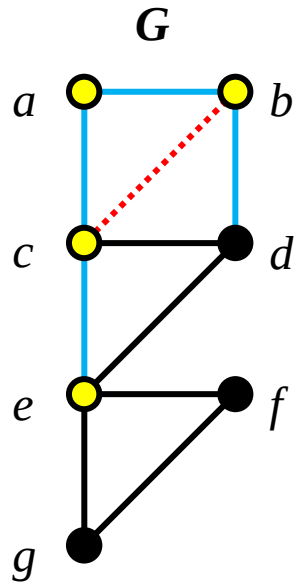


$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$
 $P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$
 $P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$
 $P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	0	1	0	0
<i>PS</i> (<i>v</i>)	0	0	0	0	0	0	0

Busca em profundidade



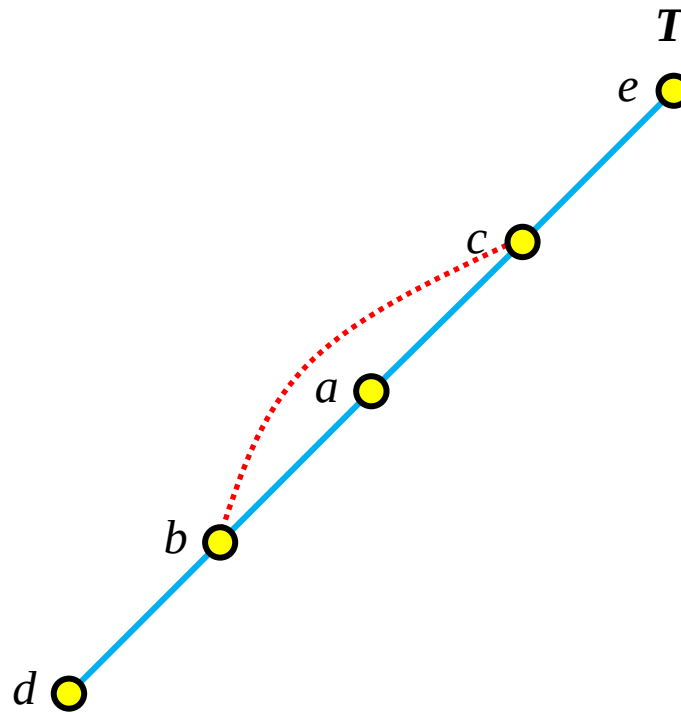
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

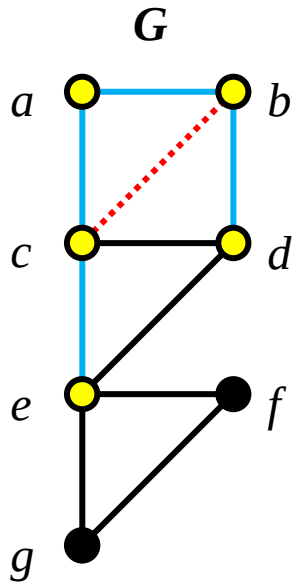
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d)$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



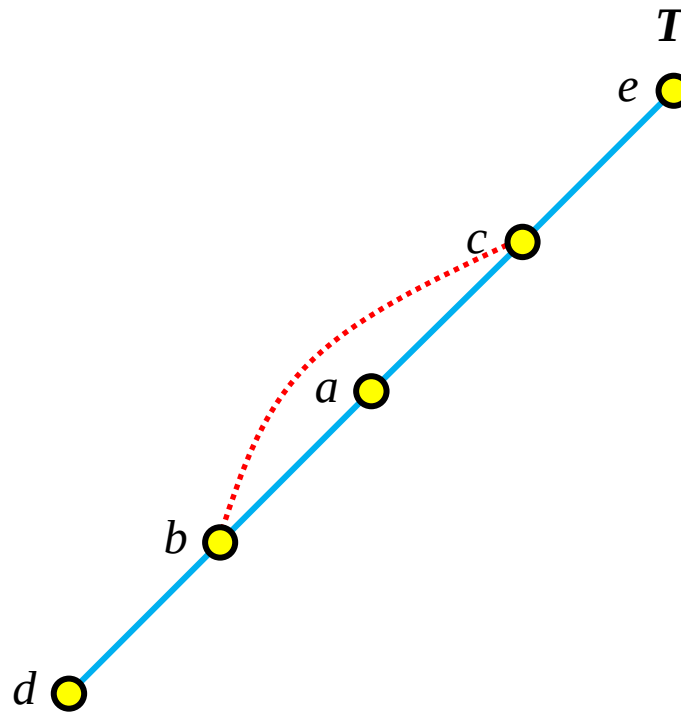
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

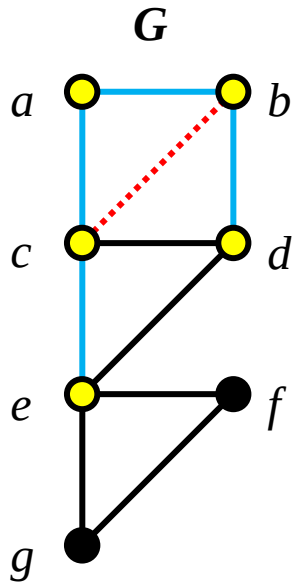
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d)$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	0	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



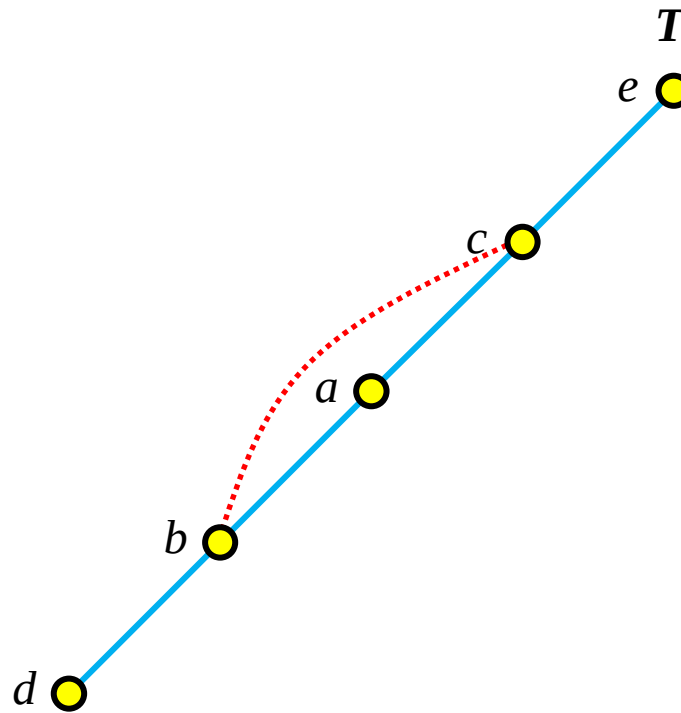
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

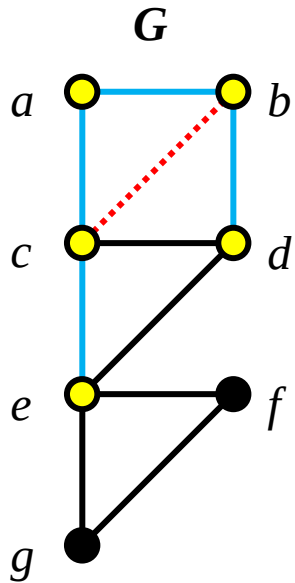
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d)$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



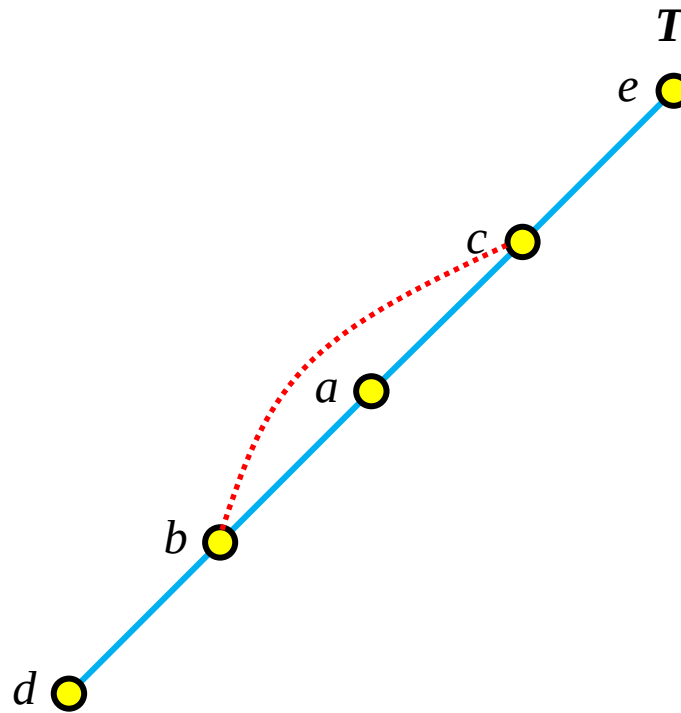
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

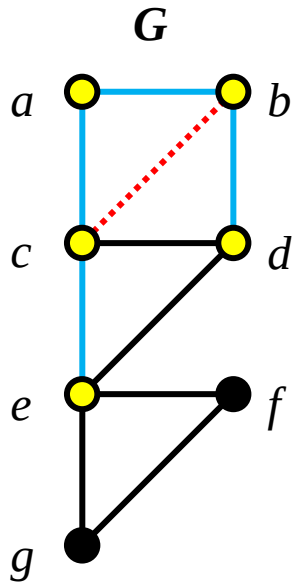
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{b, c, e\}$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



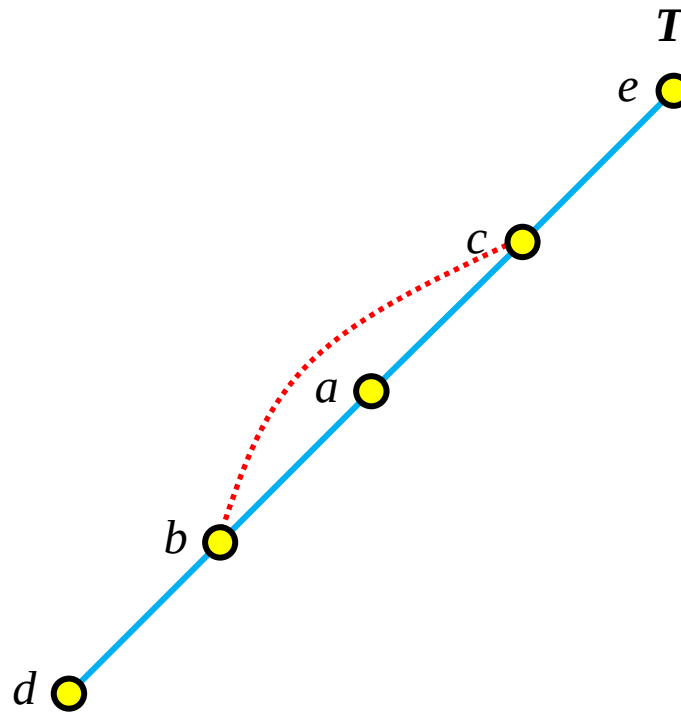
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

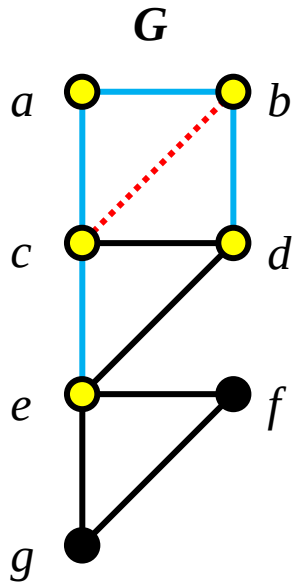
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, c, e\}$

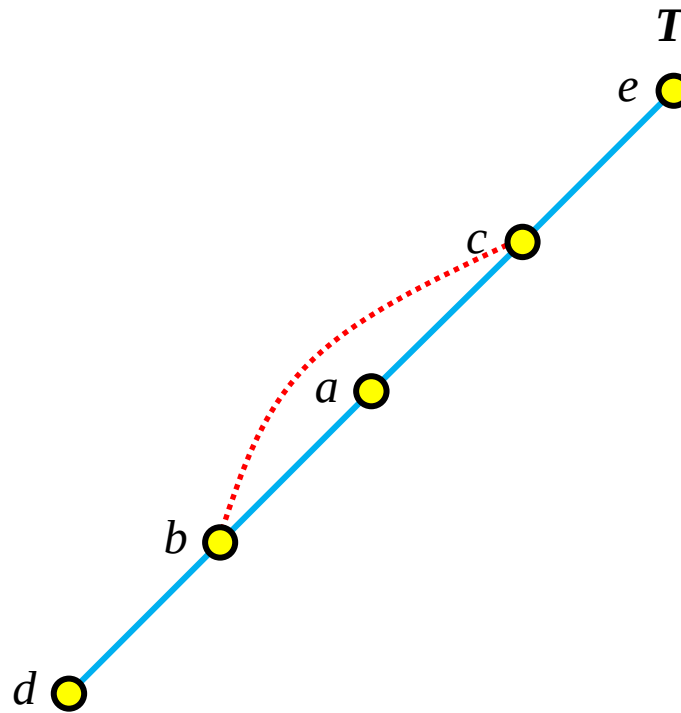


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade

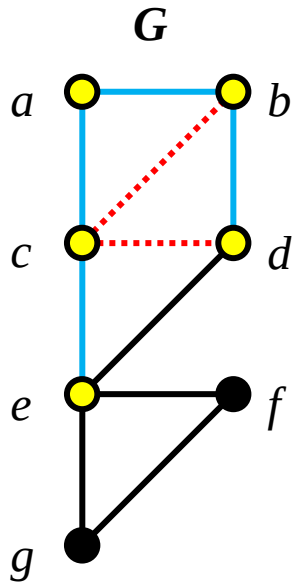


$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$
 $P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$
 $P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$
 $P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$
 $P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, e\}$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



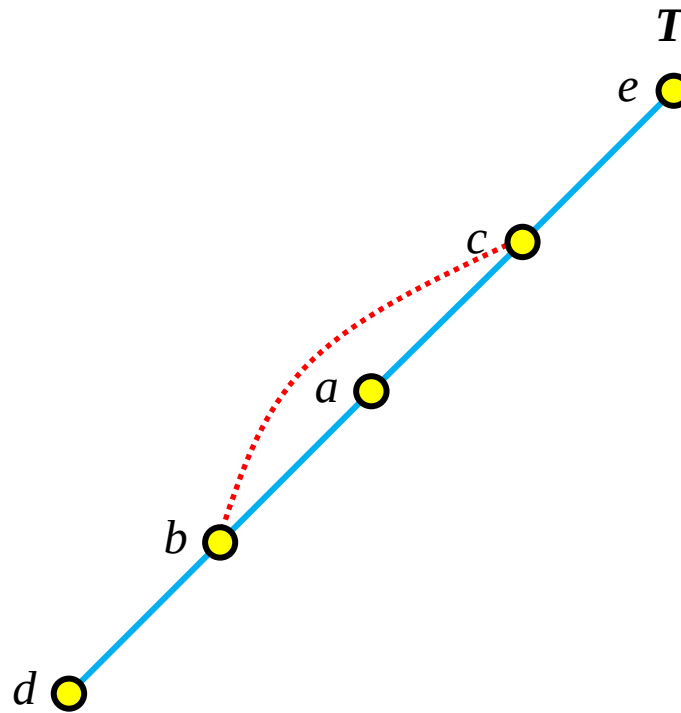
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

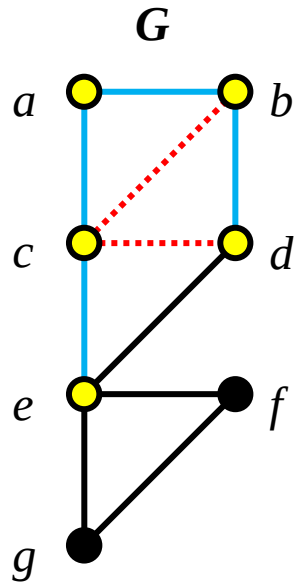
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, e\}$

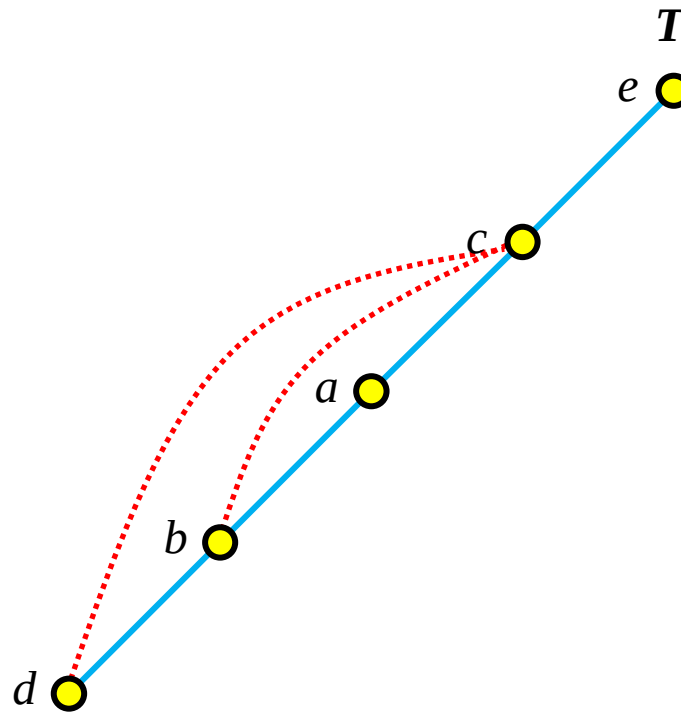


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade

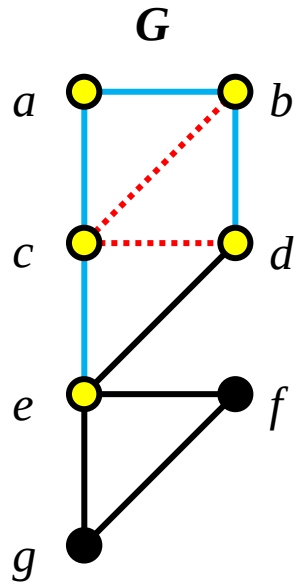


$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$
 $P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$
 $P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$
 $P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$
 $P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, e\}$

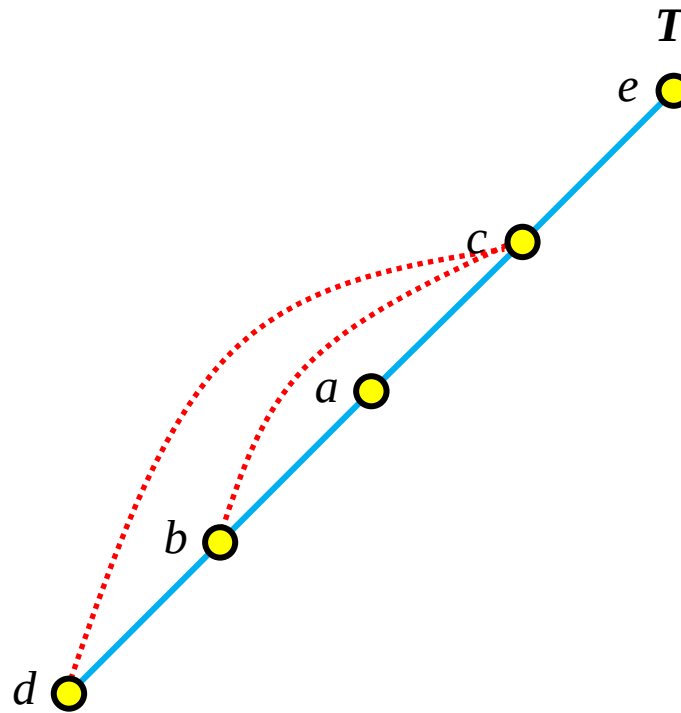


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade

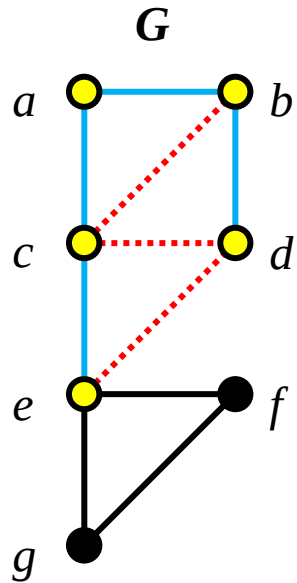


$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$
 $P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$
 $P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$
 $P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$
 $P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



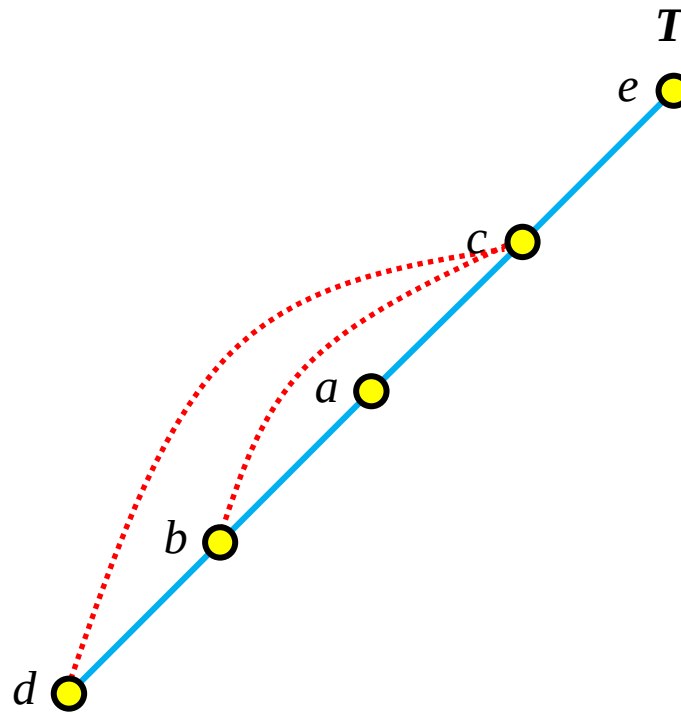
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

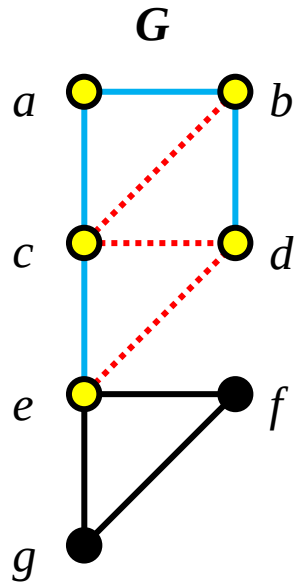
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	0	0	0	0	0	0	0

Busca em profundidade



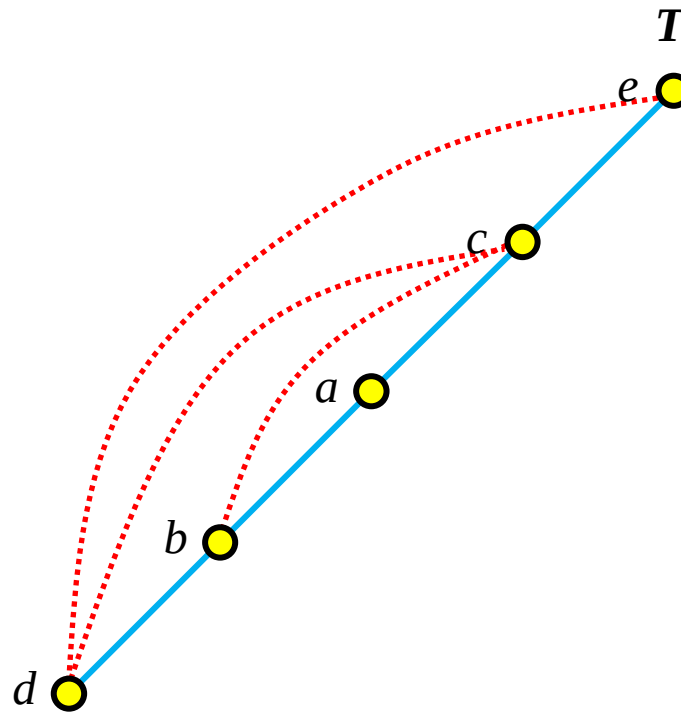
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

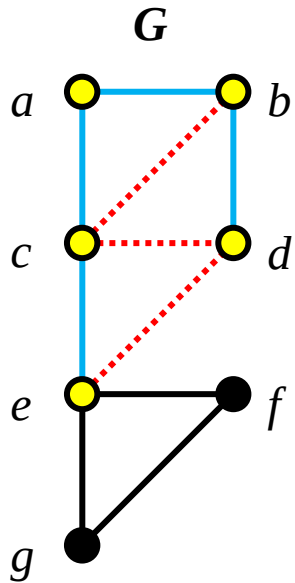
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	0	0	0	0	0	0	0

Busca em profundidade



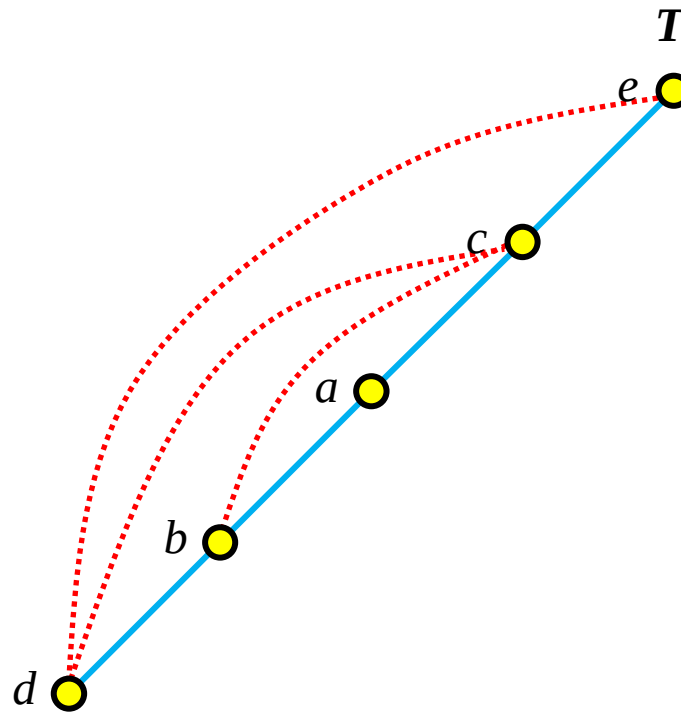
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

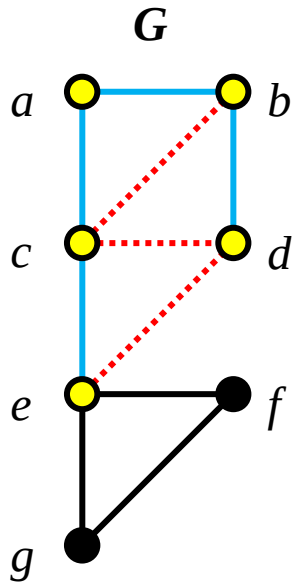
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	0	0	0	6	0	0	0

Busca em profundidade



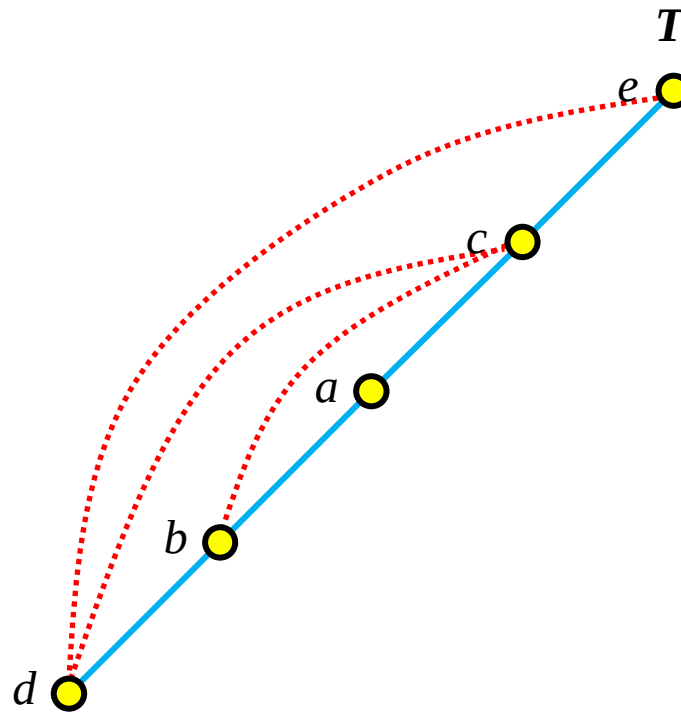
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, c\}$

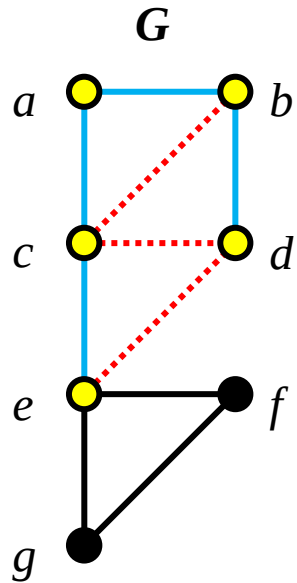
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	0	7	0	6	0	0	0

Busca em profundidade



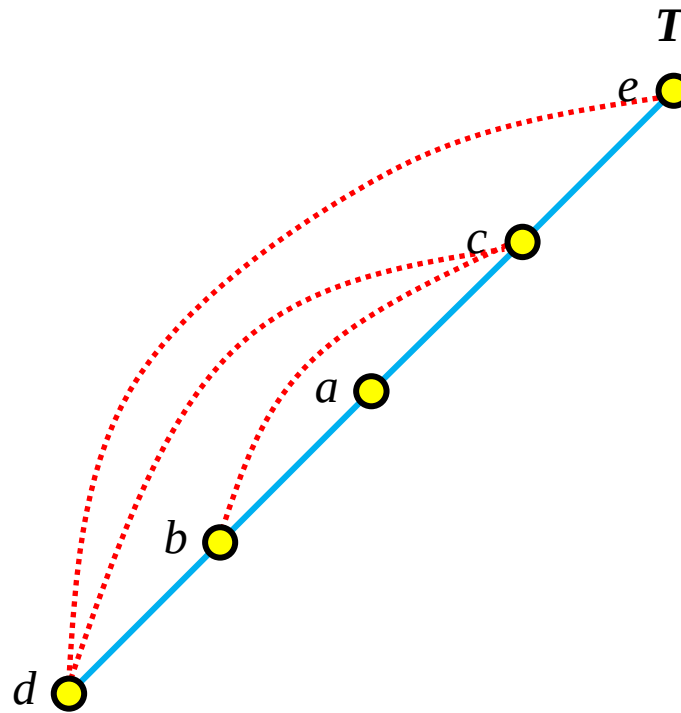
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

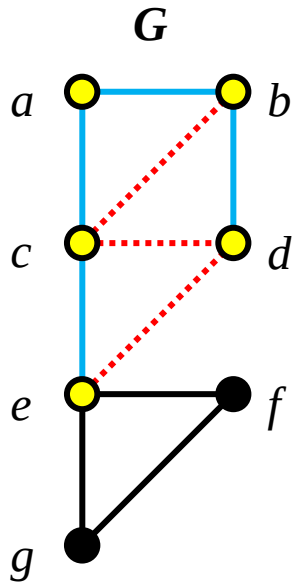
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	5	1	0	0
<i>PS</i> (<i>v</i>)	0	7	0	6	0	0	0

Busca em profundidade



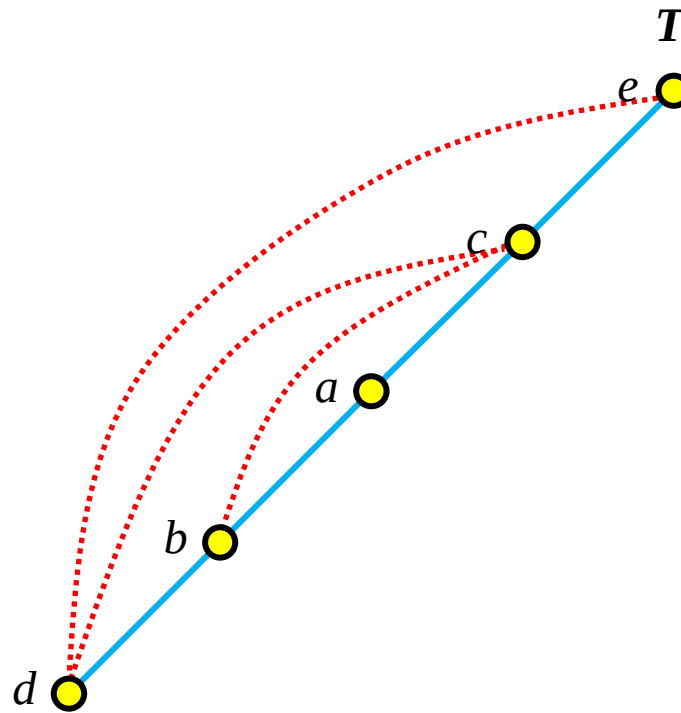
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, b, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

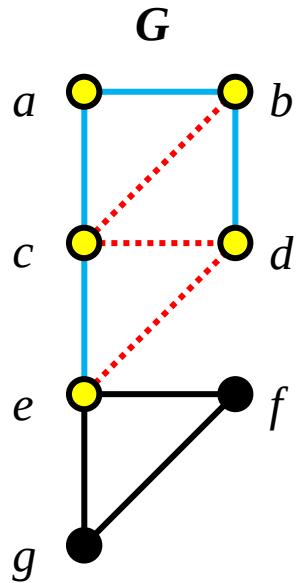
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{red}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	8	7	0	6	0	0	0

Busca em profundidade



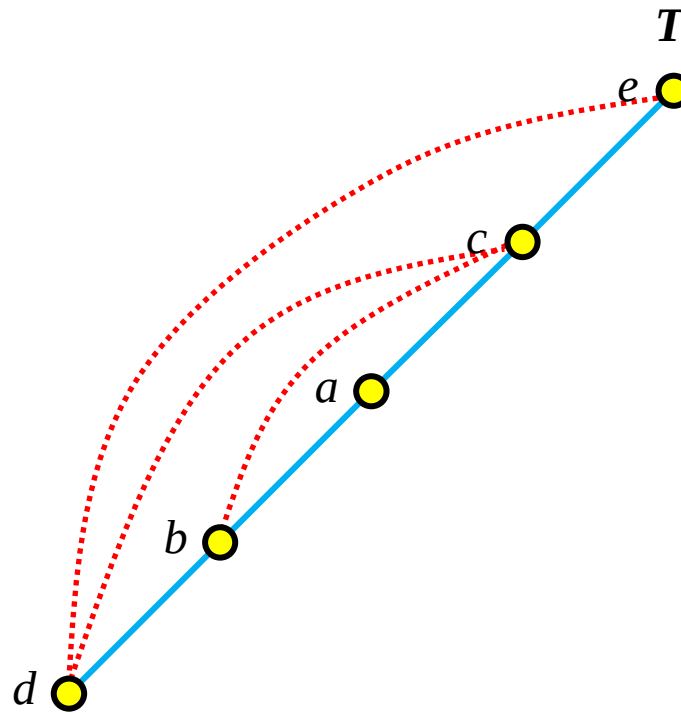
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, d, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

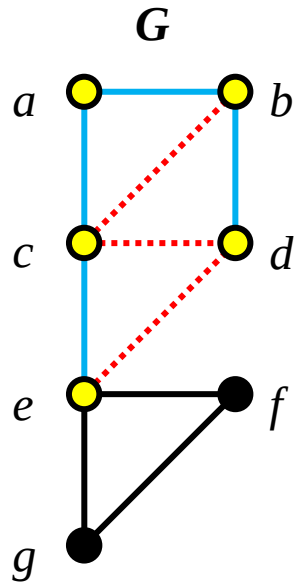
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	0	6	0	0	0

Busca em profundidade



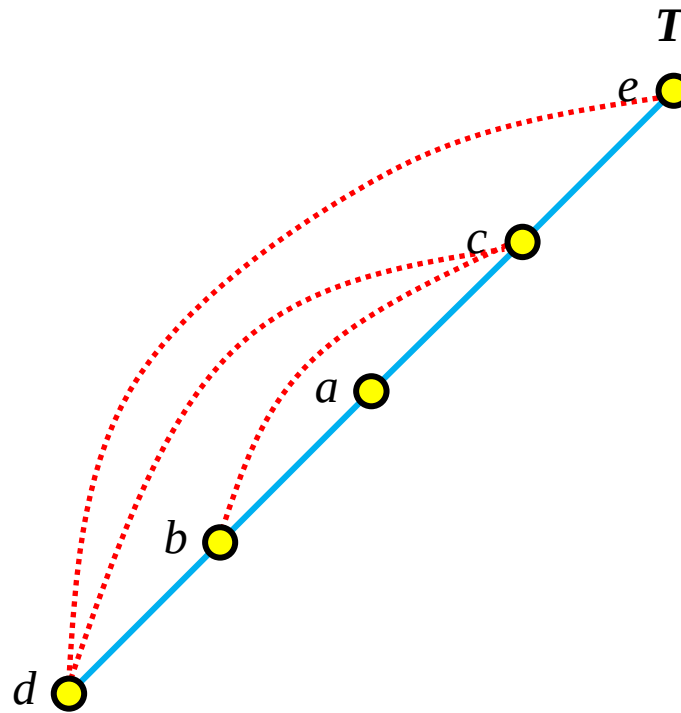
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, e\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

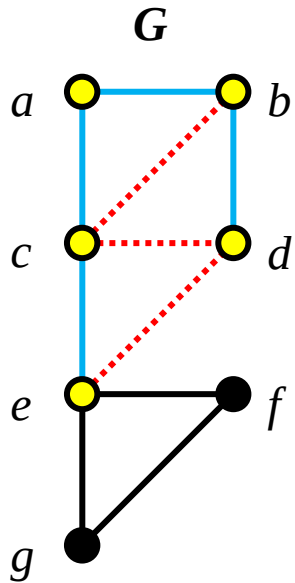
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	0	6	0	0	0

Busca em profundidade



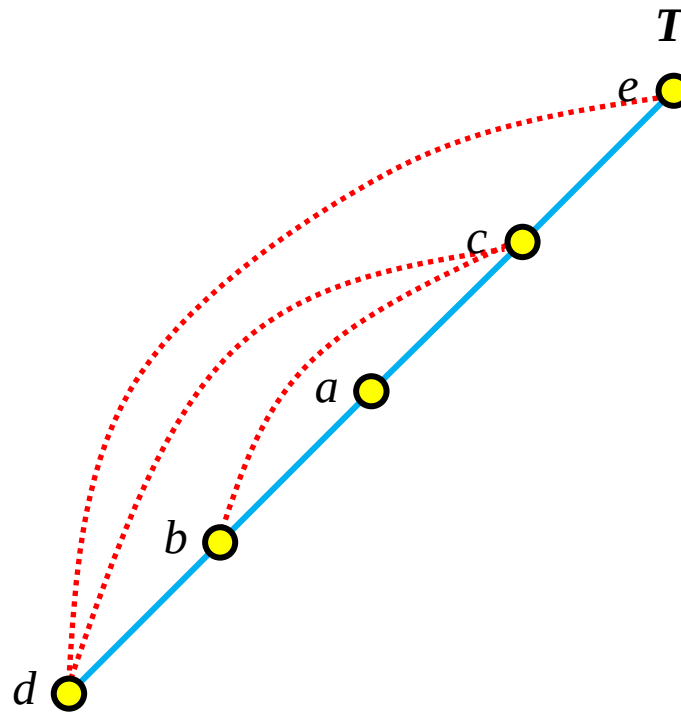
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

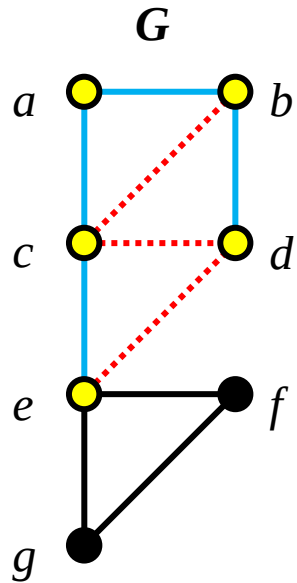
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	5	1	0	0
<i>PS</i> (<i>v</i>)	8	7	0	6	0	0	0

Busca em profundidade



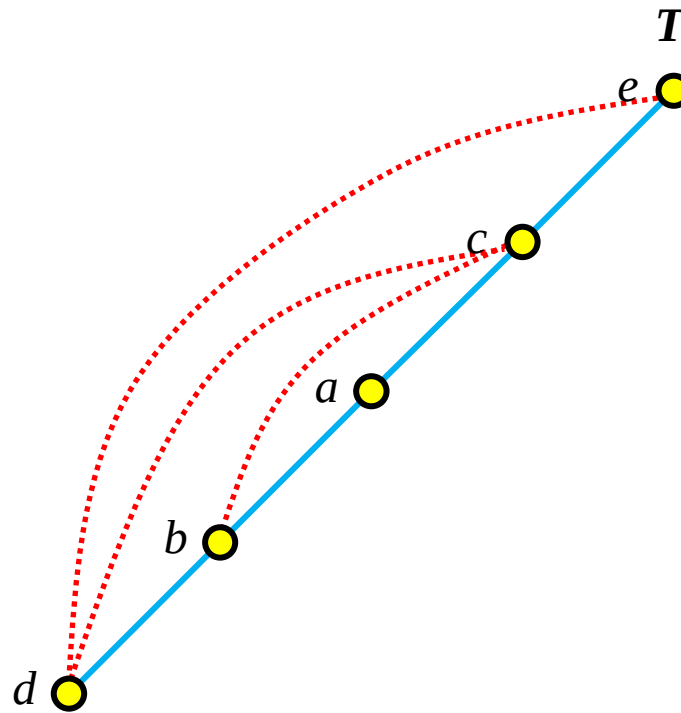
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, d, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

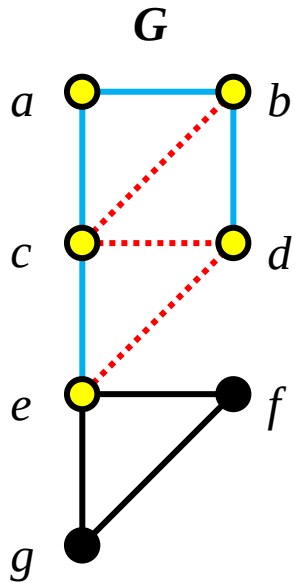
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



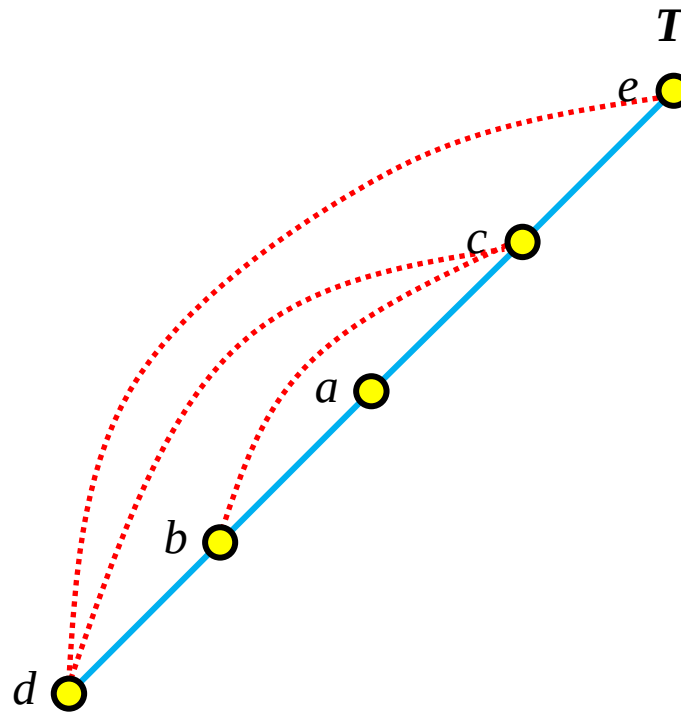
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, f, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

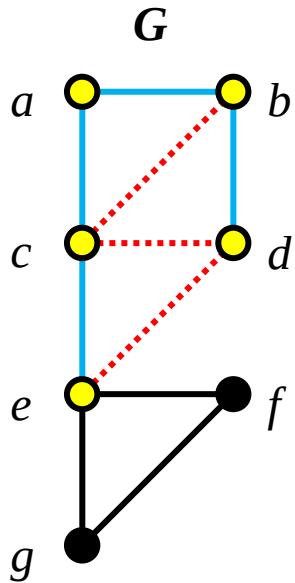
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



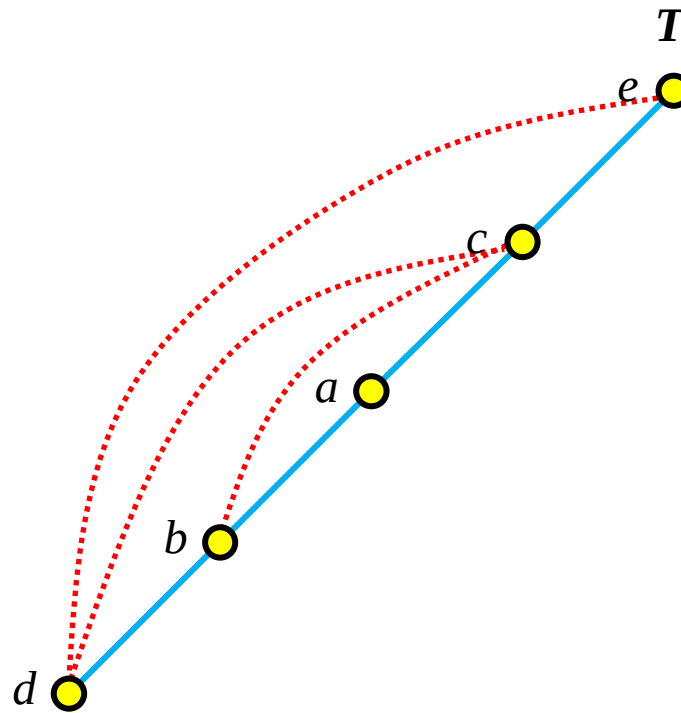
$P(e) \rightarrow N(e) = \{c, d, f, g\}$

$P(c) \rightarrow N(c) = \{a, b, d, e\}$

$P(a) \rightarrow N(a) = \{b, c\}$

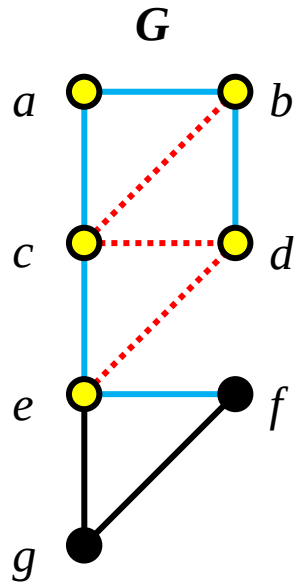
$P(b) \rightarrow N(b) = \{a, c, d\}$

$P(d) \rightarrow N(d) = \{b, c, e\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



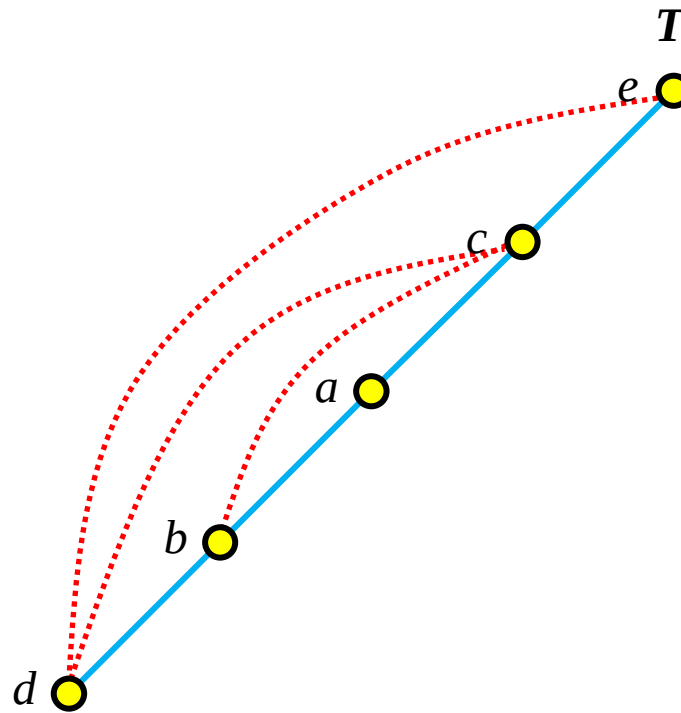
$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, \textcolor{blue}{f}, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

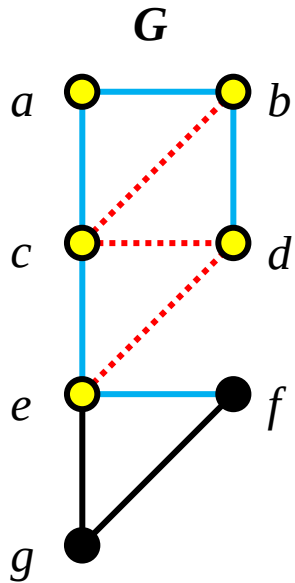
$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	0	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



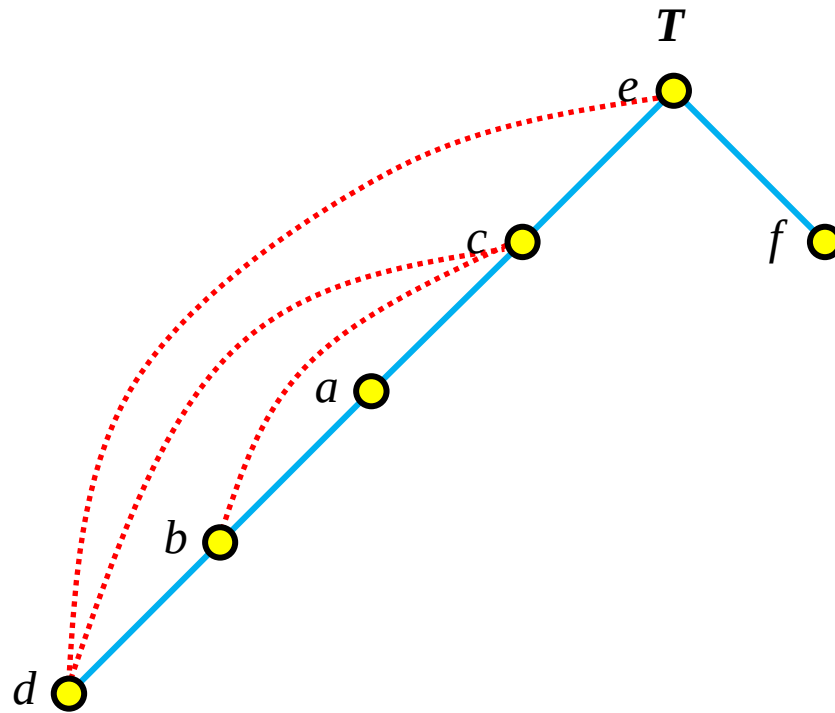
$P(e) \rightarrow N(e) = \{c, d, f, g\}$

$P(c) \rightarrow N(c) = \{a, b, d, e\}$

$P(a) \rightarrow N(a) = \{b, c\}$

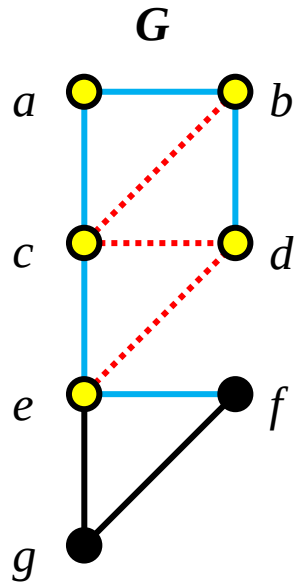
$P(b) \rightarrow N(b) = \{a, c, d\}$

$P(d) \rightarrow N(d) = \{b, c, e\}$

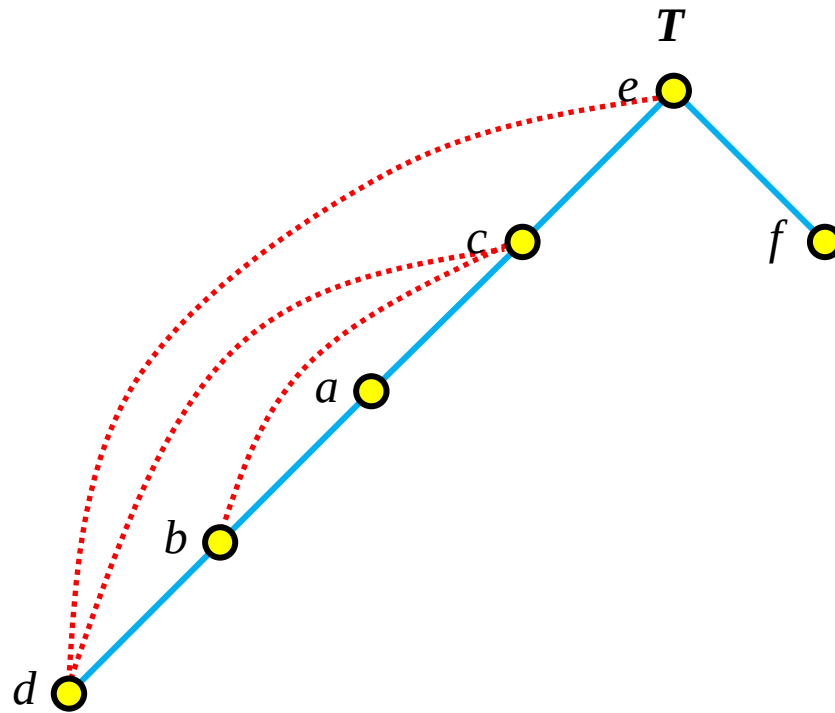


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	8	7	9	6	0	0	0

Busca em profundidade

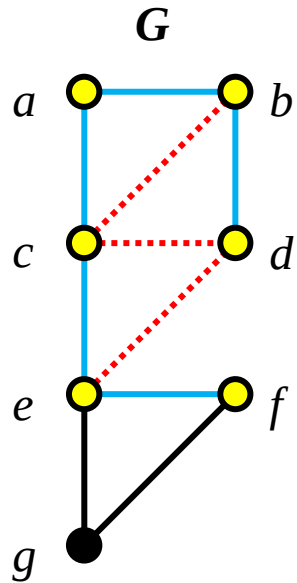


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f)$

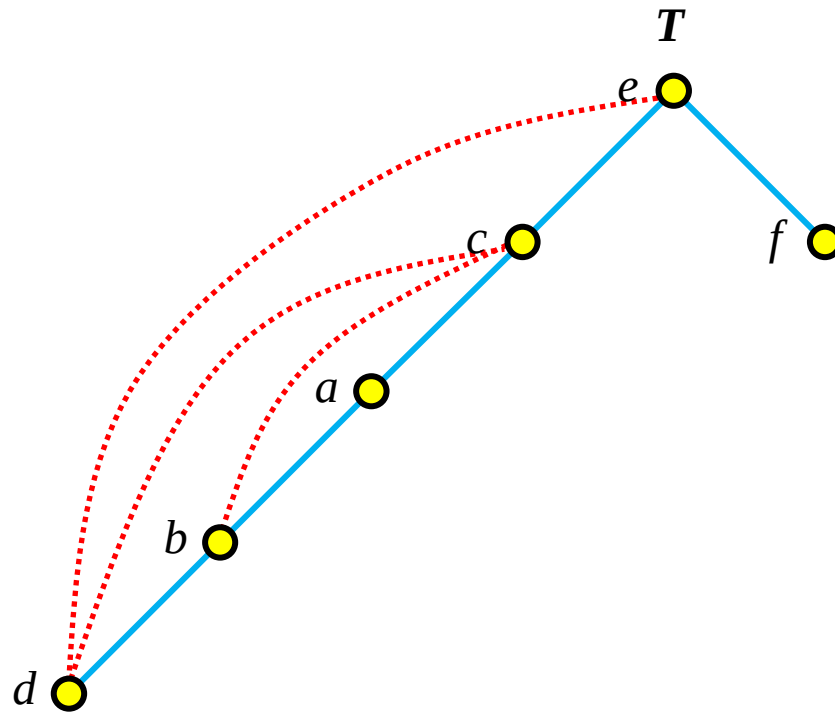


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	0	0
$PS(v)$	8	7	9	6	0	0	0

Busca em profundidade

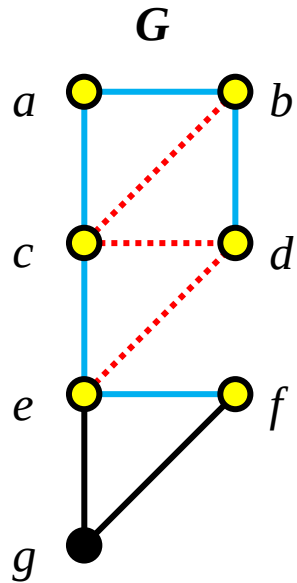


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f)$

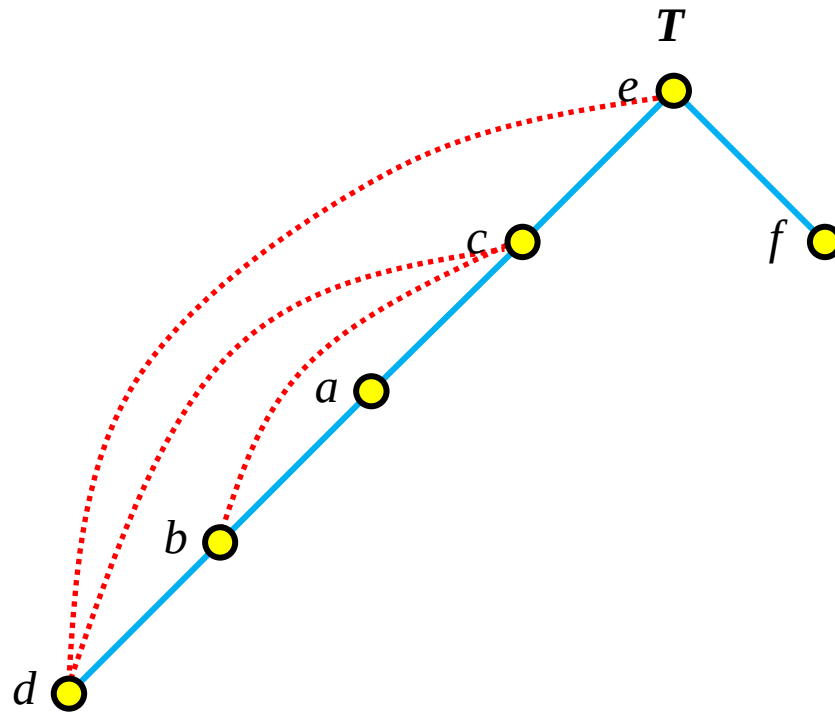


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE</i> (<i>v</i>)	3	4	2	5	1	0	0
<i>PS</i> (<i>v</i>)	8	7	9	6	0	0	0

Busca em profundidade

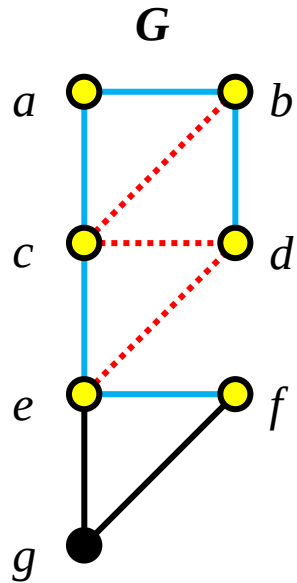


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f)$

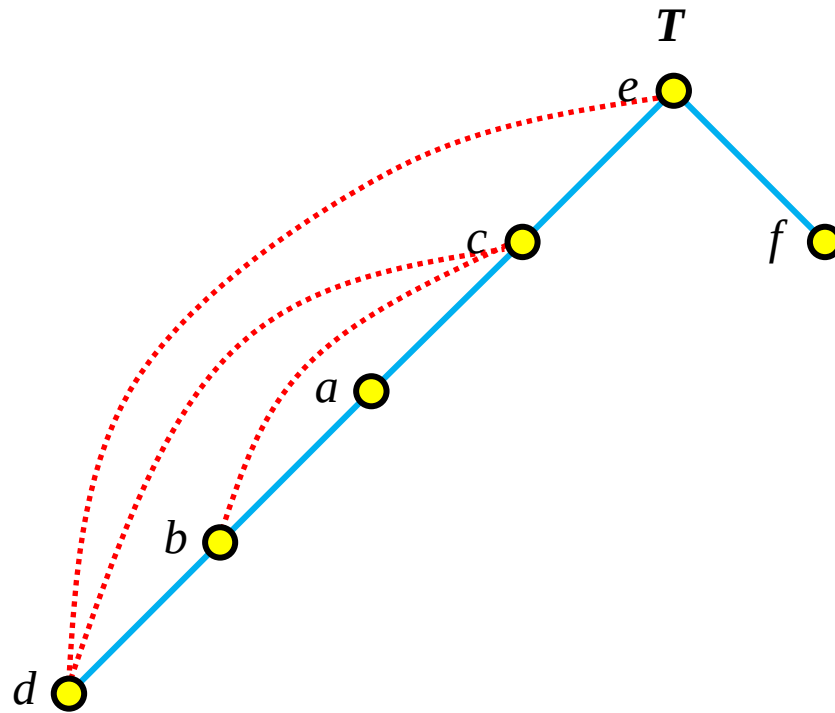


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

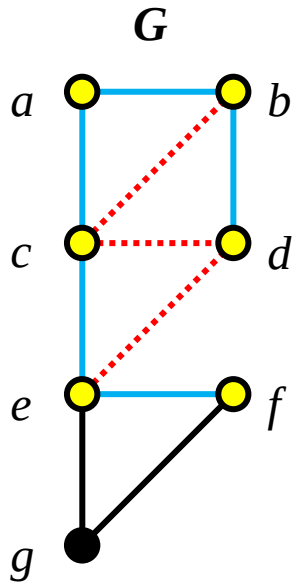


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$

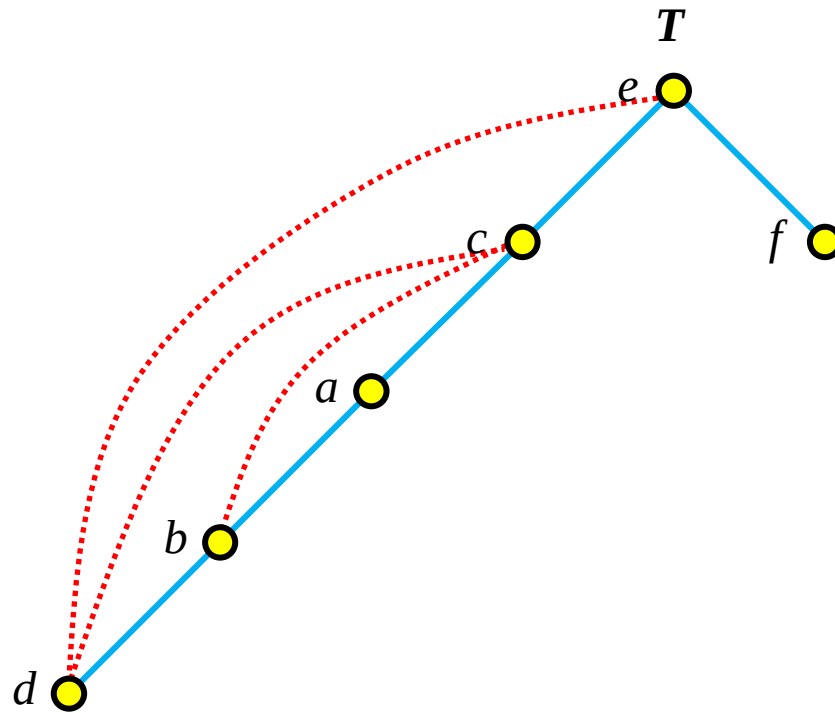


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

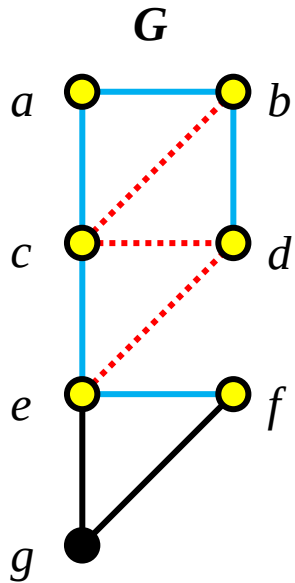


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$

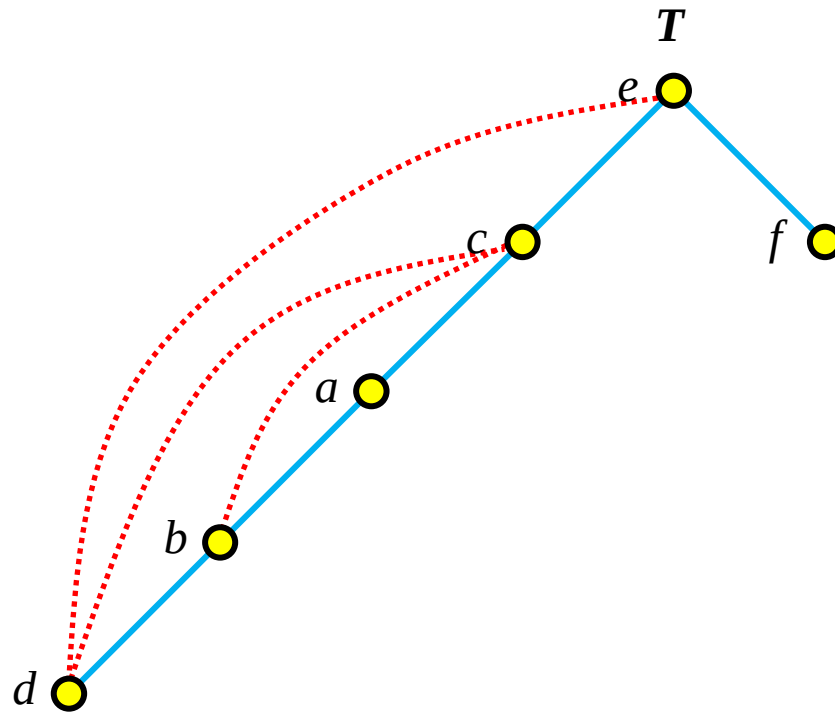


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	10	0
$PS(v)$	8	7	9	6	0	0	0

Busca em profundidade

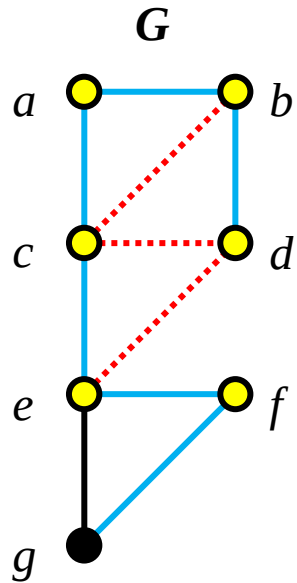


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$

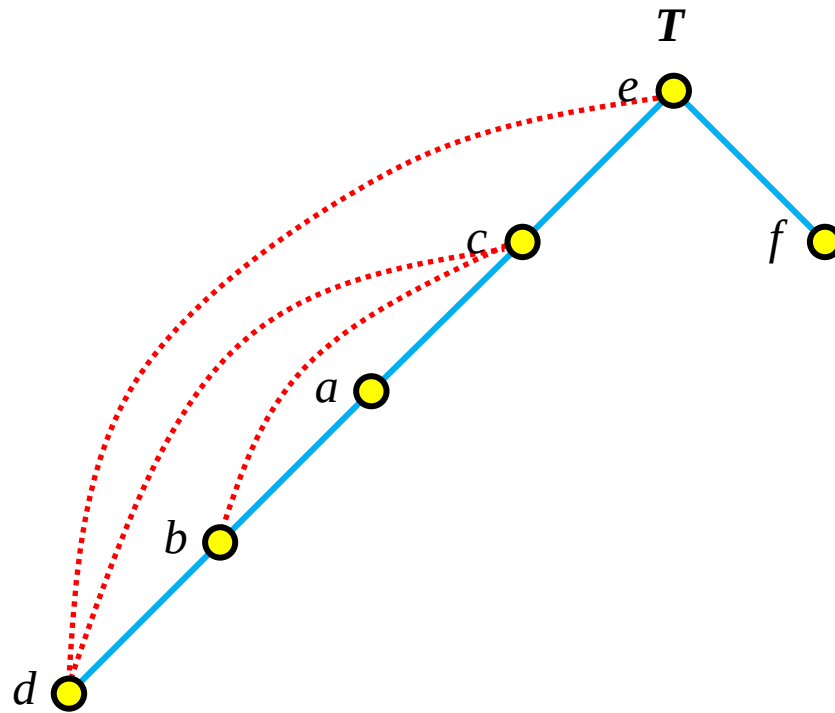


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

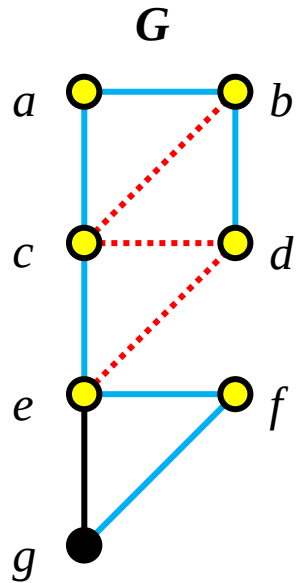


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$

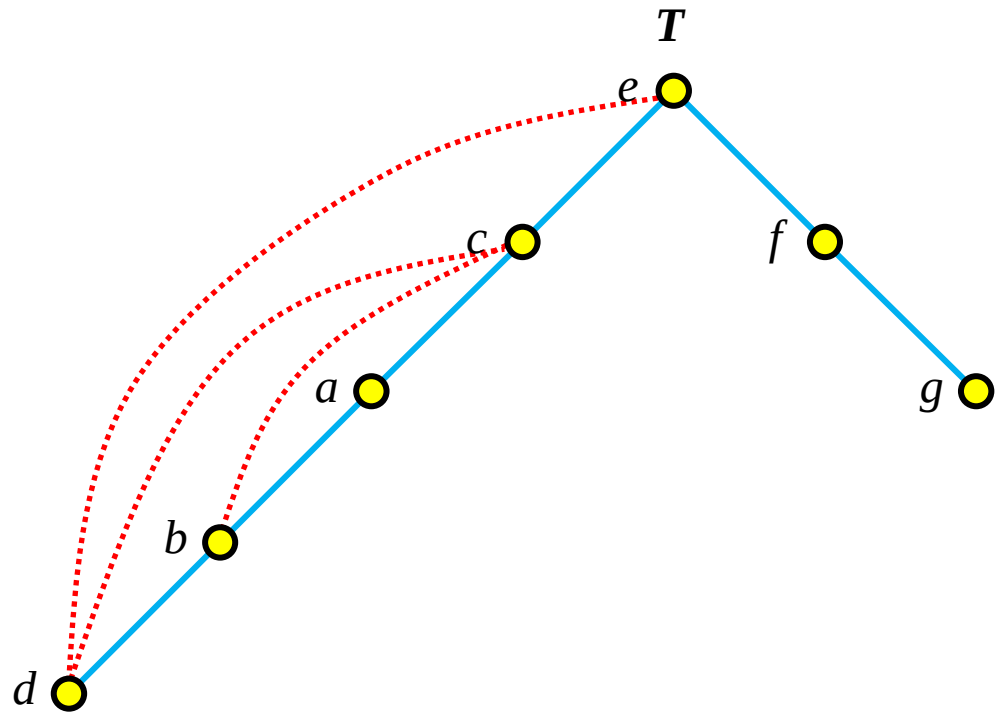


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

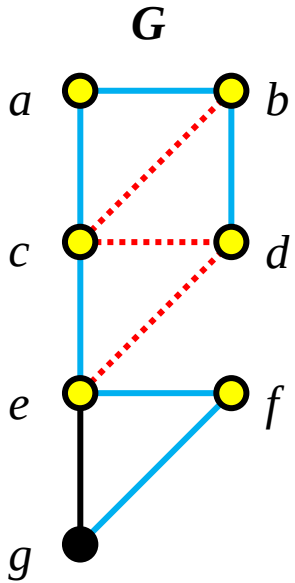


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



$$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, \textcolor{blue}{f}, g\}$$

$$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$$

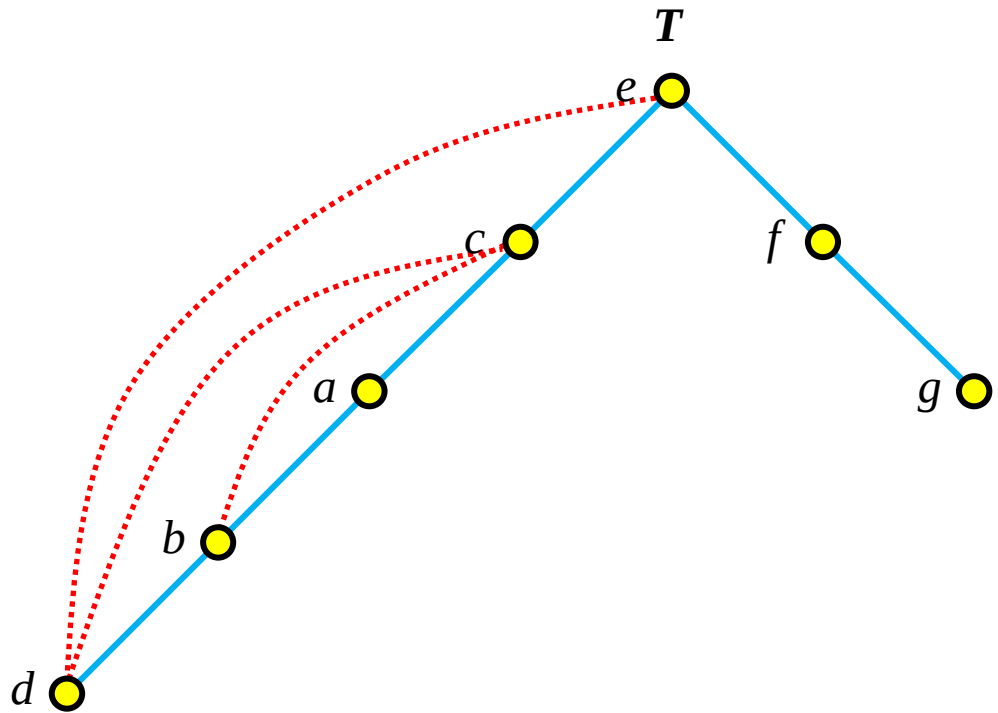
$$P(a) \rightarrow N(a) = \{\mathbf{b}, \mathbf{c}\}$$

$$P(b) \rightarrow N(b) = \{a, c, d\}$$

$$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$$

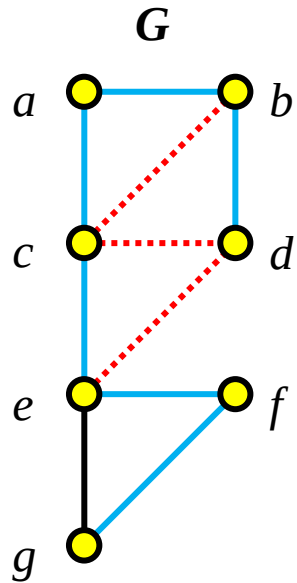
$$P(f) \rightarrow N(f) = \{\mathbf{e}, \mathbf{g}\}$$

$$P(g)$$

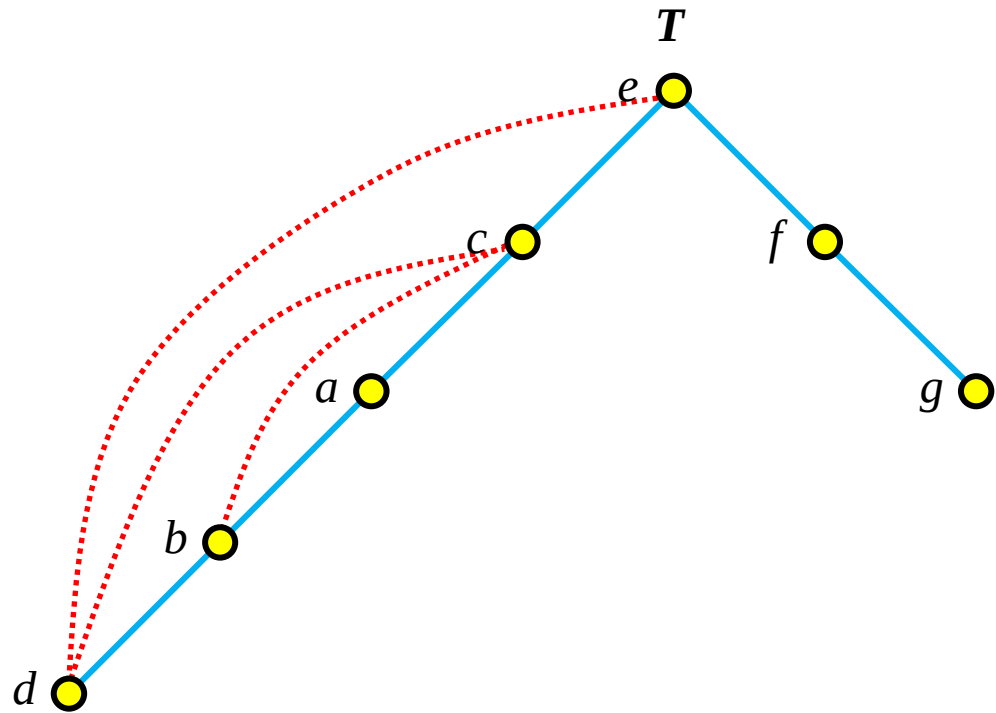


v	a	b	c	d	e	f	g
$PE(v)$	3	4	2	5	1	10	0
$PS(v)$	8	7	9	6	0	0	0

Busca em profundidade

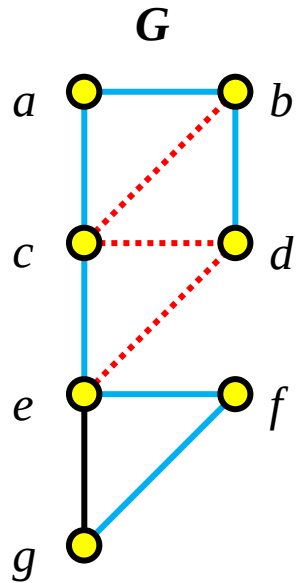


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g)$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	0
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, \textcolor{blue}{f}, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

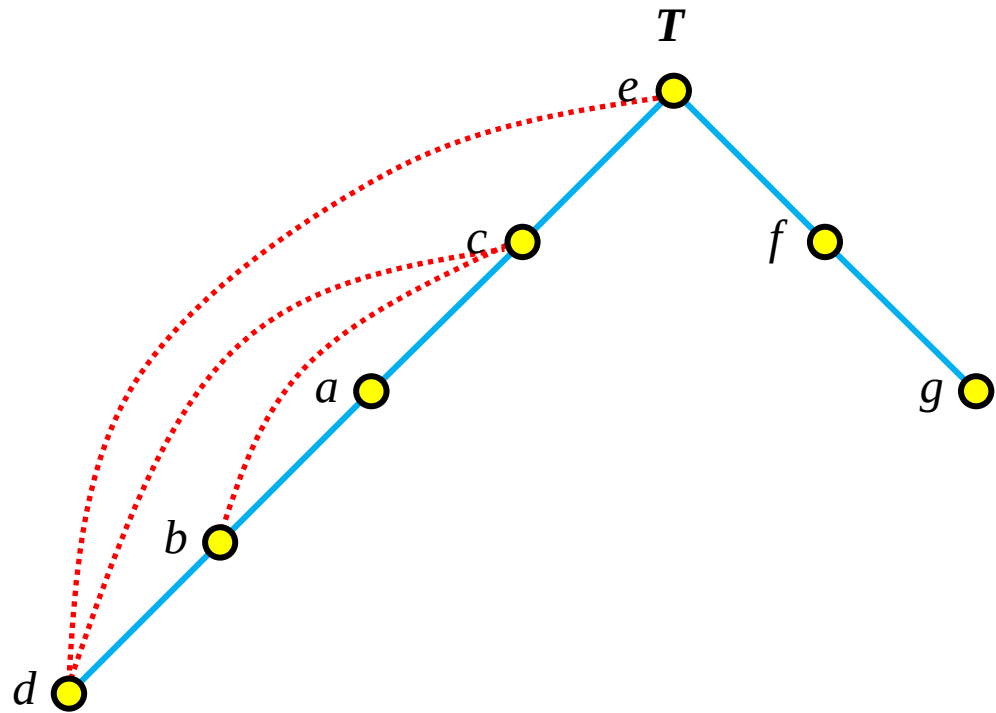
$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$

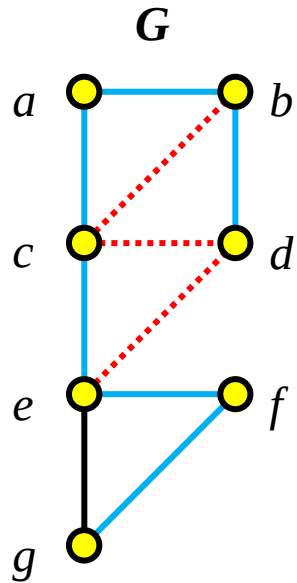
$P(f) \rightarrow N(f) = \{\textcolor{blue}{e}, g\}$

$P(g)$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
$PE(v)$	3	4	2	5	1	10	11
$PS(v)$	8	7	9	6	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{c, d, f, g\}$

$P(c) \rightarrow N(c) = \{a, b, d, e\}$

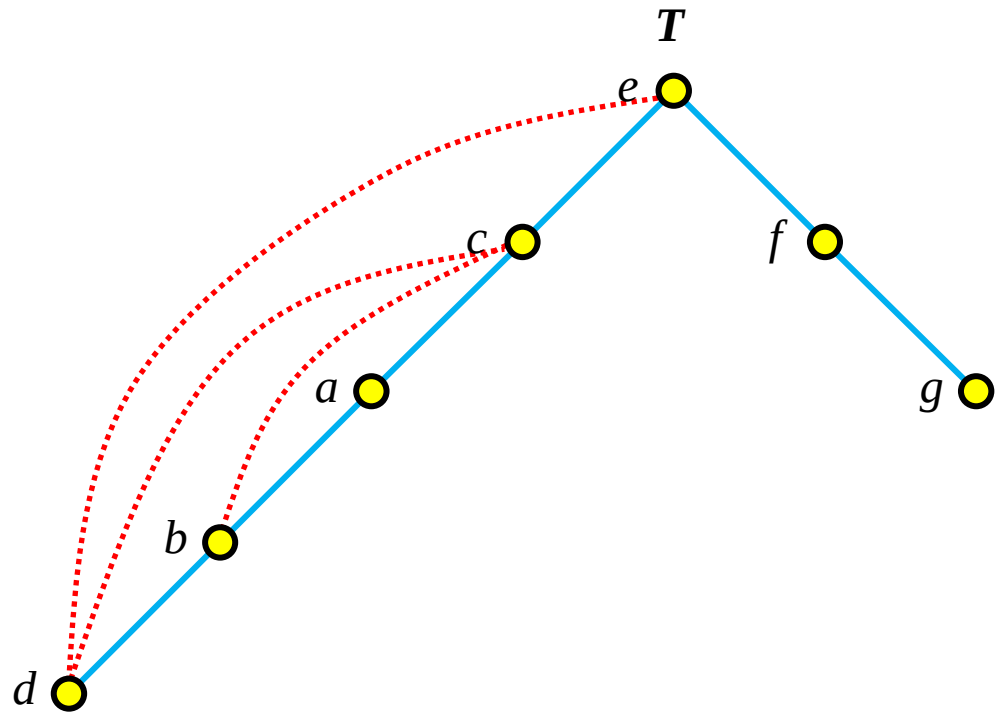
$P(a) \rightarrow N(a) = \{b, c\}$

$P(b) \rightarrow N(b) = \{a, c, d\}$

$P(d) \rightarrow N(d) = \{b, c, e\}$

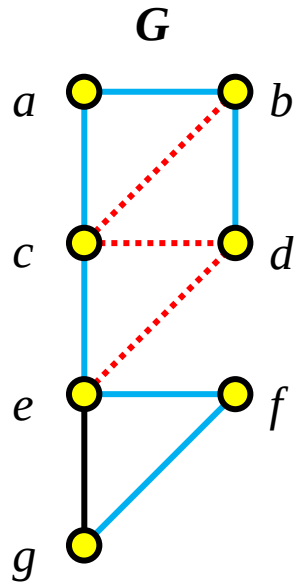
$P(f) \rightarrow N(f) = \{e, g\}$

$P(g) \rightarrow N(g) = \{e, f\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{c, d, f, g\}$

$P(c) \rightarrow N(c) = \{a, b, d, e\}$

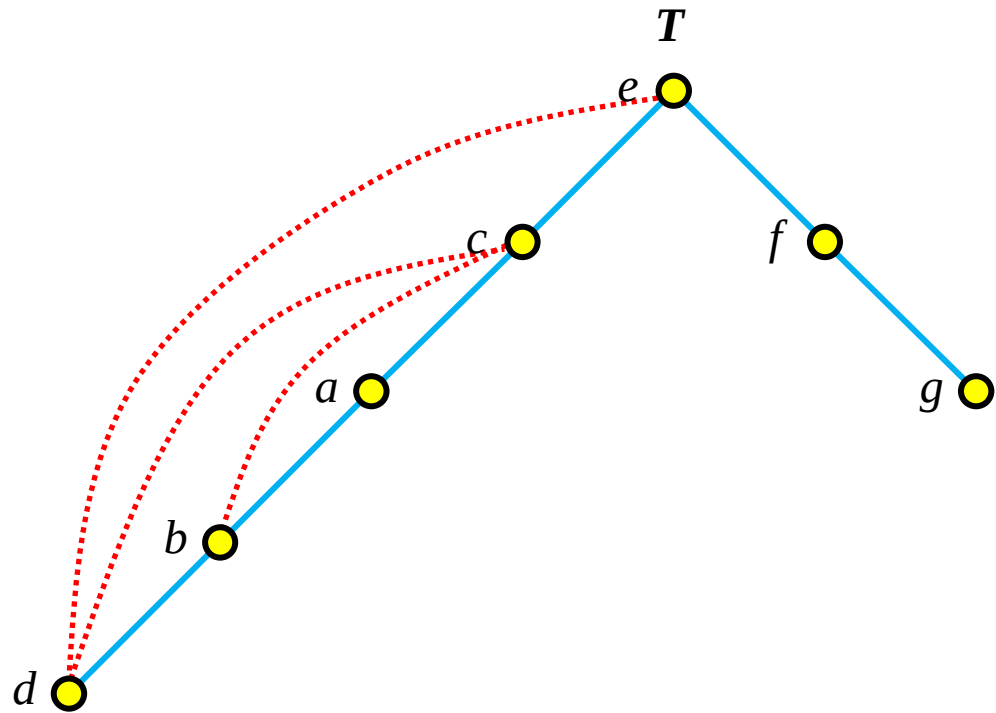
$P(a) \rightarrow N(a) = \{b, c\}$

$P(b) \rightarrow N(b) = \{a, c, d\}$

$P(d) \rightarrow N(d) = \{b, c, e\}$

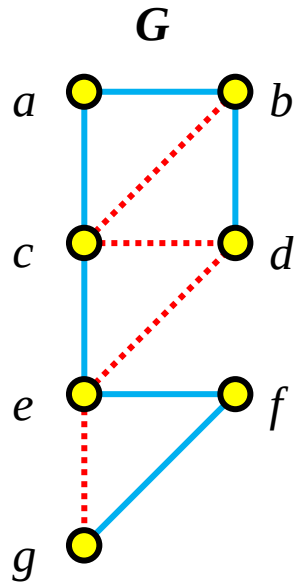
$P(f) \rightarrow N(f) = \{e, g\}$

$P(g) \rightarrow N(g) = \{e, f\}$

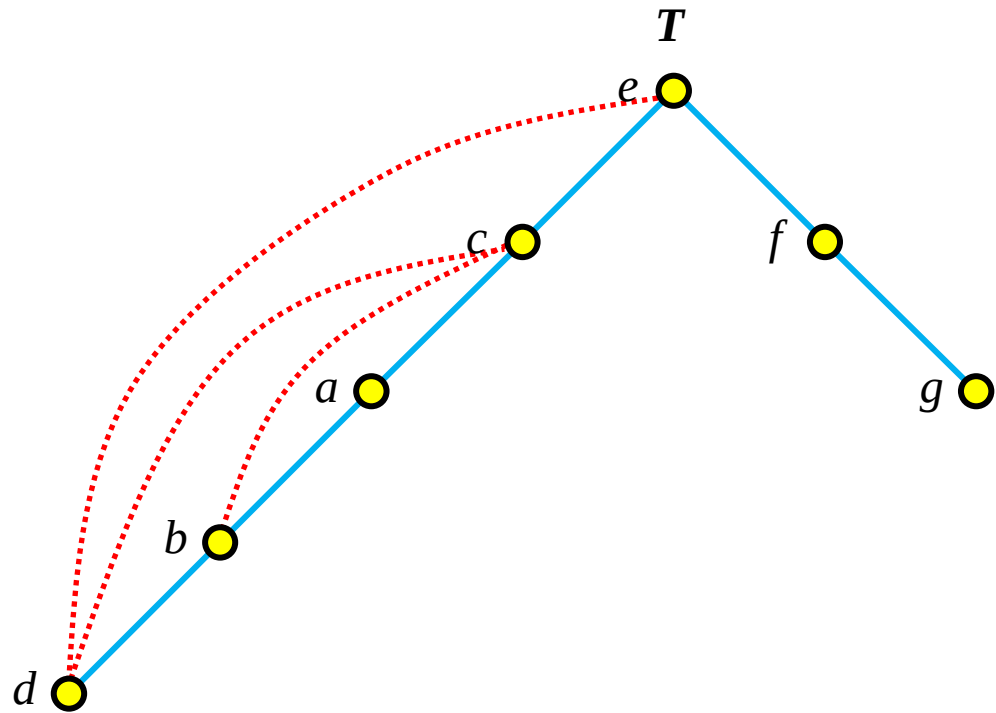


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

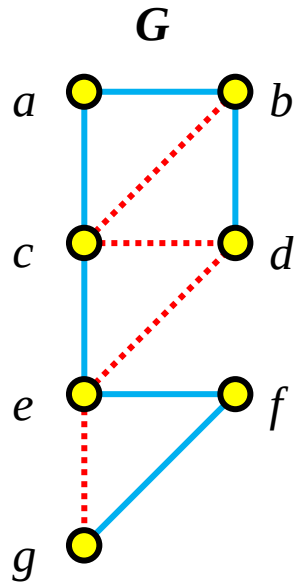


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g) \rightarrow N(g) = \{e, f\}$

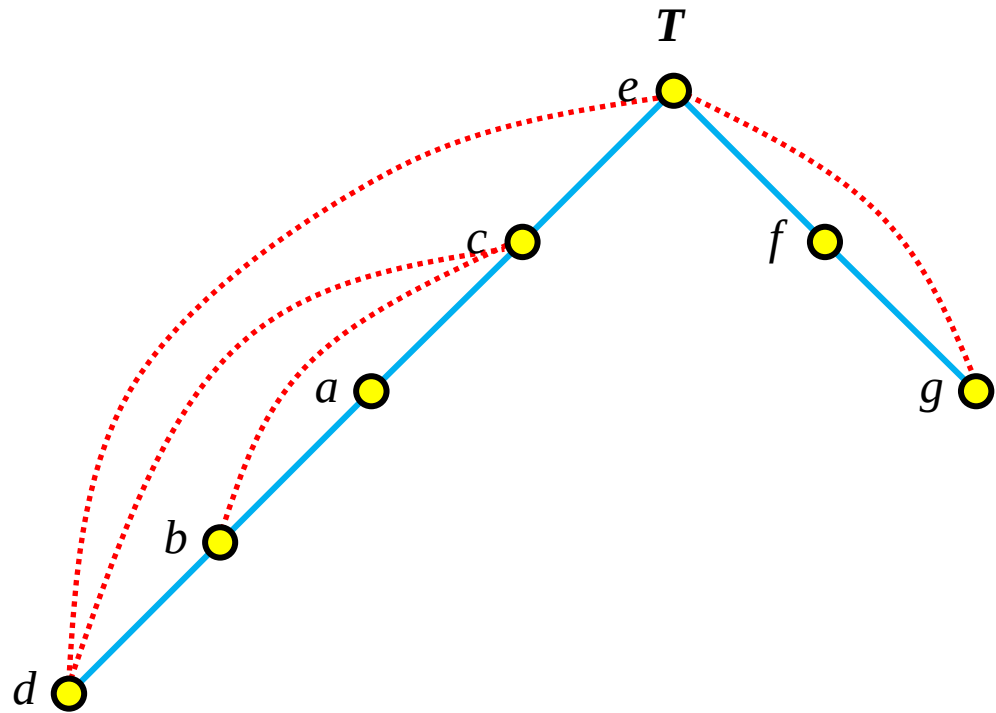


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

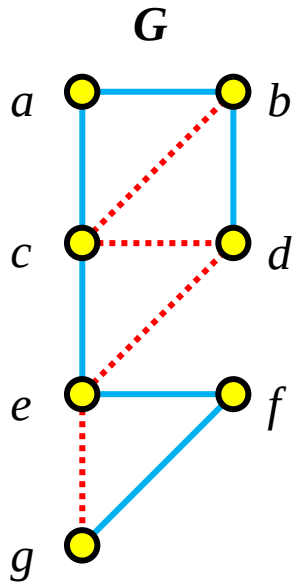


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g) \rightarrow N(g) = \{e, f\}$

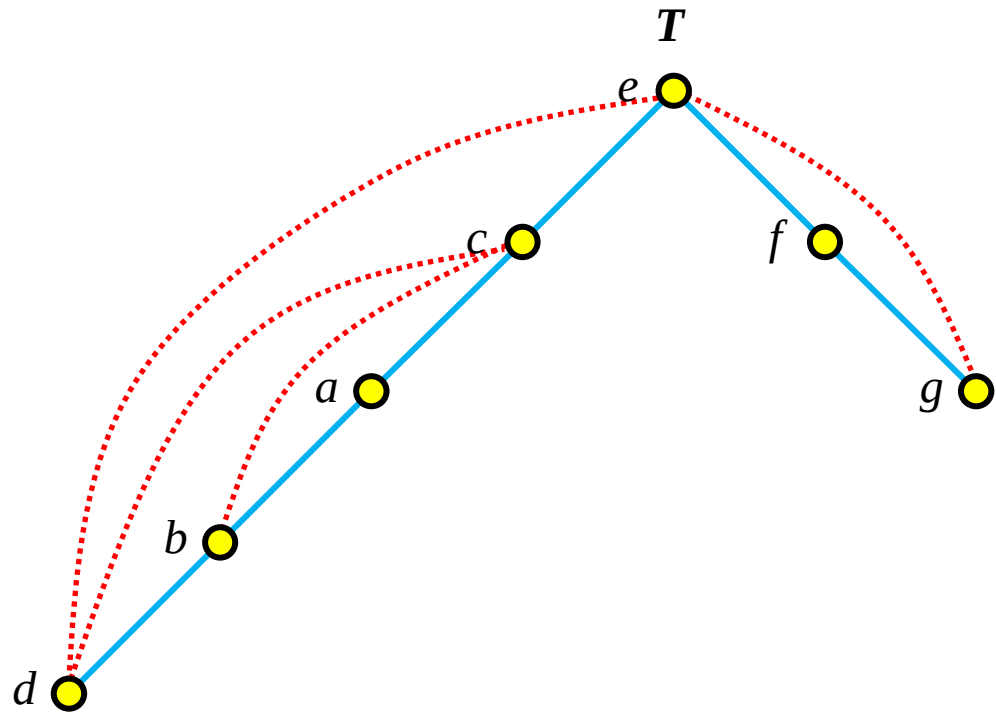


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade

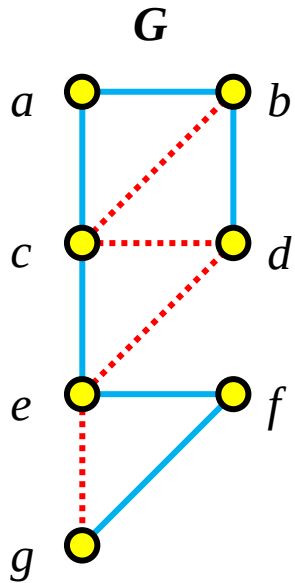


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g) \rightarrow N(g) = \{e, f\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	0

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, \textcolor{blue}{f}, g\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

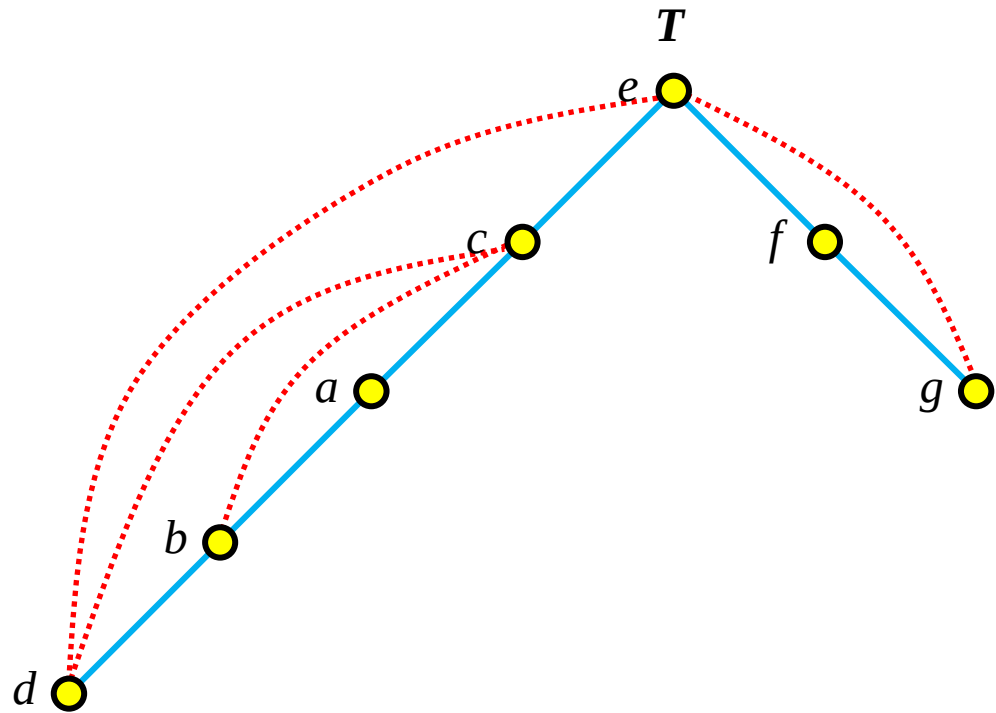
$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$

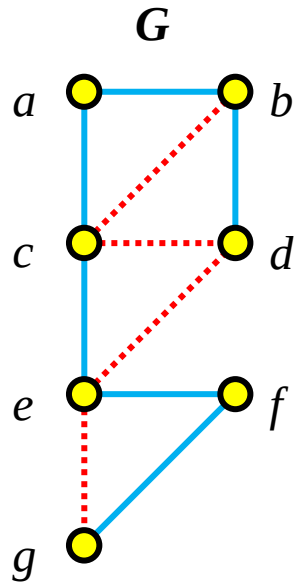
$P(f) \rightarrow N(f) = \{\textcolor{blue}{e}, g\}$

$P(g) \rightarrow N(g) = \{\textcolor{red}{e}, \textcolor{blue}{f}\}$

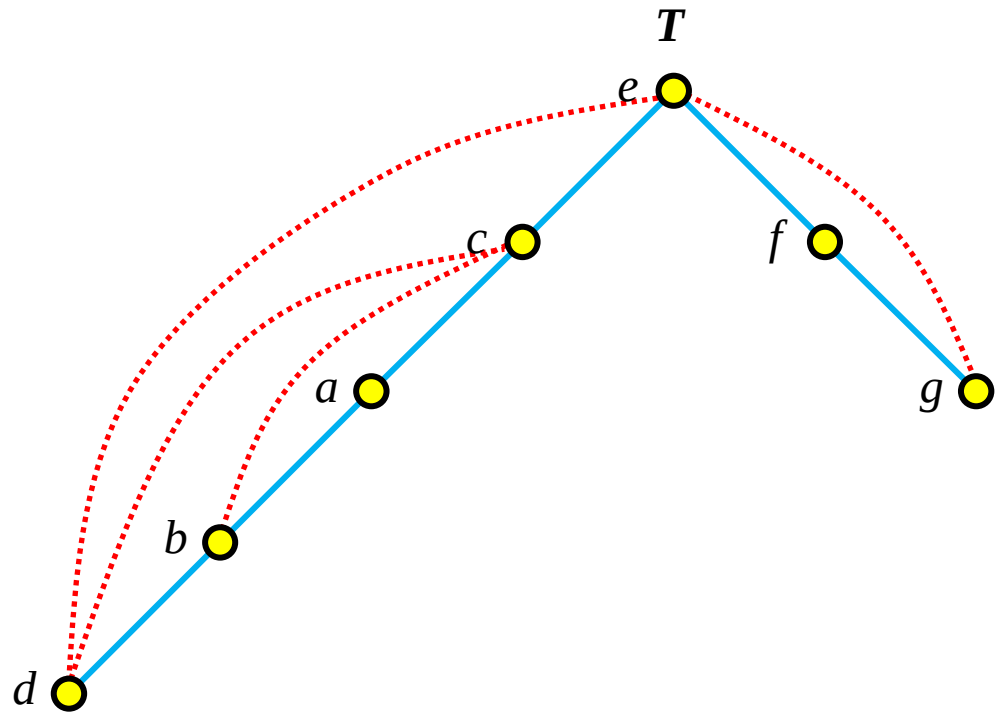


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	0	12

Busca em profundidade

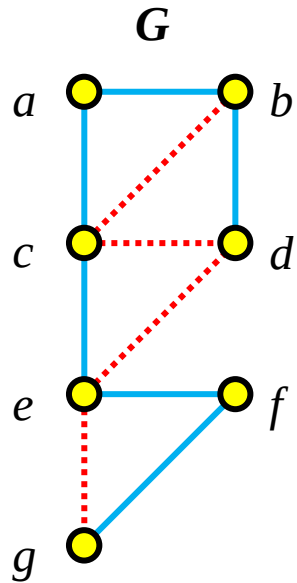


$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g) \rightarrow N(g) = \{e, f\}$



v	a	b	c	d	e	f	g
PE(v)	3	4	2	5	1	10	11
PS(v)	8	7	9	6	0	13	12

Busca em profundidade



$P(e) \rightarrow N(e) = \{\textcolor{blue}{c}, \textcolor{red}{d}, \textcolor{blue}{f}, \textcolor{red}{g}\}$

$P(c) \rightarrow N(c) = \{\textcolor{blue}{a}, \textcolor{red}{b}, \textcolor{red}{d}, \textcolor{blue}{e}\}$

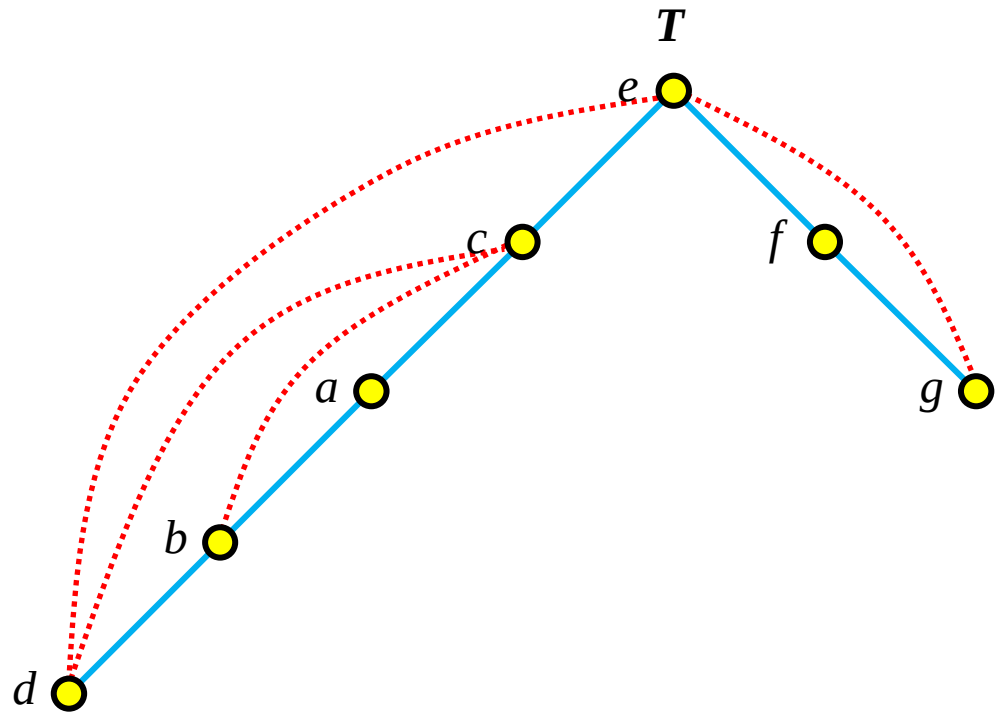
$P(a) \rightarrow N(a) = \{\textcolor{blue}{b}, \textcolor{blue}{c}\}$

$P(b) \rightarrow N(b) = \{\textcolor{blue}{a}, \textcolor{red}{c}, \textcolor{blue}{d}\}$

$P(d) \rightarrow N(d) = \{\textcolor{blue}{b}, \textcolor{red}{c}, \textcolor{red}{e}\}$

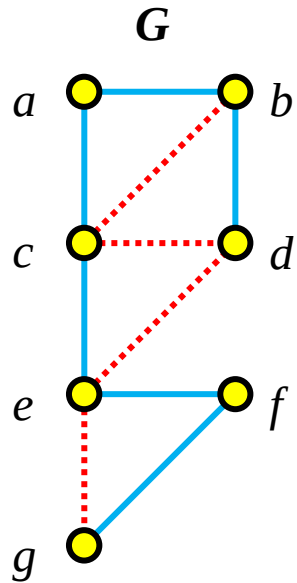
$P(f) \rightarrow N(f) = \{\textcolor{blue}{e}, \textcolor{blue}{g}\}$

$P(g) \rightarrow N(g) = \{\textcolor{red}{e}, \textcolor{blue}{f}\}$

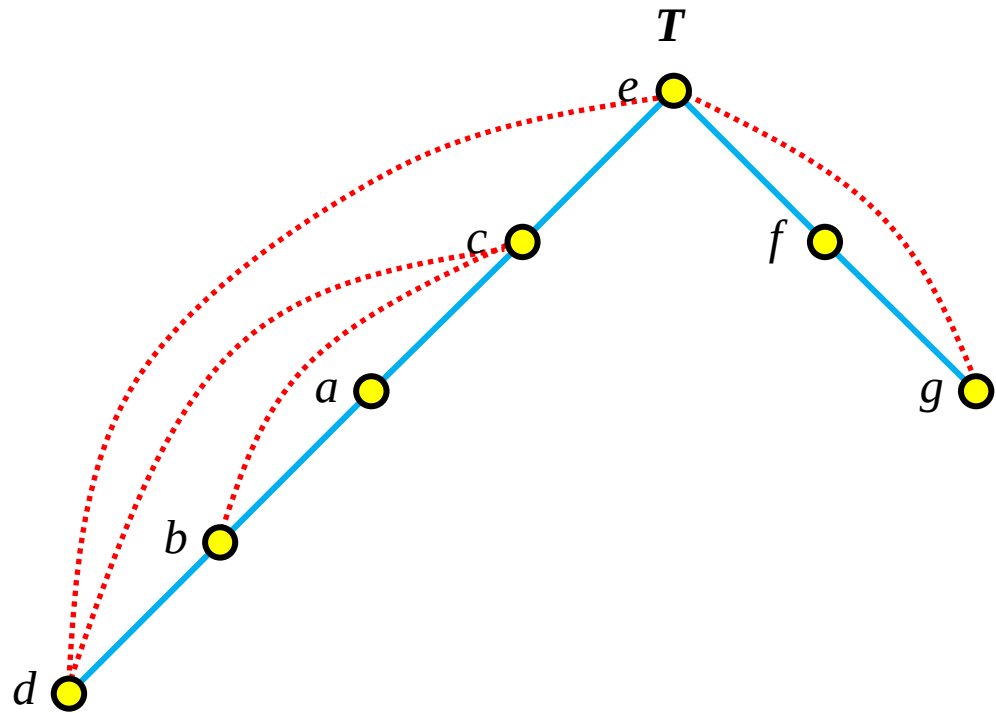


<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	0	13	12

Busca em profundidade



$P(e) \rightarrow N(e) = \{c, d, f, g\}$
 $P(c) \rightarrow N(c) = \{a, b, d, e\}$
 $P(a) \rightarrow N(a) = \{b, c\}$
 $P(b) \rightarrow N(b) = \{a, c, d\}$
 $P(d) \rightarrow N(d) = \{b, c, e\}$
 $P(f) \rightarrow N(f) = \{e, g\}$
 $P(g) \rightarrow N(g) = \{e, f\}$



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
<i>PE(v)</i>	3	4	2	5	1	10	11
<i>PS(v)</i>	8	7	9	6	14	13	12

Busca em profundidade

“Esgotar o filho antes de esgotar o pai”

“A próxima aresta a ser visitada parte sempre do vértice **mais recente** na busca”

Busca em profundidade

Complexidade da busca em profundidade:

$$O(n + m)$$

(onde $n = |V(G)|$ e $m = |E(G)|$)

Busca em profundidade

Se o grafo for desconexo, a busca alcançará apenas os vértices que estão conectados ao vértice-raiz da busca por caminhos!!

Alternativa:

modificar a chamada externa para alcançar todo o grafo

*enquanto existe v em $V(G)$ tal que $PE(v) = 0$ faça
executar $P(v)$*